Topics in Mathematics Science VII
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Introduction to Koszul Duality

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Convention

Throughout the course, $\Bbbk$ is a field. All rings are unital and associative. All modules are right modules unless otherwise specified. When we discuss dimensions, we mean dimensions as $\Bbbk$-vector spaces.

Pale green passages with a dark green left rule contain supplementary material; pale beige passages with a red left rule contain advanced material and may be skipped by beginners.

1 Graded algebras, modules, and duals

1.1 Algebras and modules

Definition 1.1. *A $\Bbbk$-algebra is a $\Bbbk$-vector space with an associative unital multiplication that is bilinear, meaning linear in each variable.*

A right A-module is understood to be a $\Bbbk$-vector space M with a bilinear action satisfying $(ma)b = m(ab)$ and $m1_A = m$ for $m \in M$ and $a, b \in A$. A module homomorphism $f : M \to N$ is linear and satisfies $f(ma) = f(m)a$. A homomorphism of $\Bbbk$-algebras is a linear map preserving multiplication and the identity.

Example 1.2. *(1) The matrix ring $\mathrm{Mat}_n(\Bbbk)$ is a $\Bbbk$-algebra. The space of $1 \times n$ row vectors is a right $\mathrm{Mat}_n(\Bbbk)$-module by matrix multiplication.*

(2) Let $\Bbbk A_n$ be the upper triangular matrix ring, which is also a $\Bbbk$-algebra.

$$\Bbbk A_n = \begin{pmatrix} \Bbbk & \Bbbk & \cdots & \Bbbk \\ 0 & \Bbbk & \cdots & \Bbbk \\ 0 & 0 & \ddots & \vdots \\ 0 & \cdots & 0 & \Bbbk \end{pmatrix}$$

For those familiar with path algebras, $\Bbbk A_n$ is the path algebra of a linearly oriented A_n quiver, which explains the notation. We use matrix calculations, so no knowledge of quivers is required. Write e_{ij} for the matrix with 1 in entry (i,j) and zero elsewhere. Our convention composes paths from left to right: e_{ij} represents $i \to j$, and $e_{ij}e_{j\ell} = e_{i\ell}$.

1.2 Categories and functors

We organise modules together with their homomorphisms. A category records both: specifying the objects alone does not specify a category.

Definition 1.3. *A category $\mathcal{C}$ consists of objects, a set $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ of morphisms $f : X \to Y$ for each pair of objects, and composition sending $f : X \to Y$ and $g : Y \to Z$ to $g \circ f : X \to Z$. Each object X has an identity $\mathrm{id}_X : X \to X$, and*

$$h \circ (g \circ f) = (h \circ g) \circ f, \qquad \mathrm{id}_Y \circ f = f = f \circ \mathrm{id}_X \quad \text{for all composable } f, g, h.$$

A morphism $f : X \to Y$ is an isomorphism if some $g : Y \to X$ satisfies $g \circ f = \mathrm{id}_X$ and $f \circ g = \mathrm{id}_Y$; we then write $X \cong Y$. A full subcategory of $\mathcal{C}$ consists of chosen objects of $\mathcal{C}$, with all morphisms between them and the same composition.

Example 1.4. *For a $\Bbbk$-algebra A, we use the notation*

$$\begin{aligned} \operatorname{Mod} A &:= \textit{category of right } A\textit{-modules and } A\textit{-linear maps},\\ \operatorname{mod} A &:= \textit{full subcategory of finitely generated right } A\textit{-modules}. \end{aligned}$$

In either category, $\mathrm{Hom}_A(M,N)$ is the space of A-linear maps $M \to N$; composition is the usual composition of maps. For $A = \Bbbk$, these are the categories of all and of finite-dimensional vector spaces, respectively. More generally, if $\dim_{\Bbbk} A < \infty$, an A-module is finitely generated if and only if it is finite-dimensional: a finite generating set gives a quotient of $A^{\oplus n}$, and a finite $\Bbbk$-basis is also an A-generating set.

Definition 1.5. *A functor $F : \mathcal{C} \to \mathcal{D}$ assigns an object $F(X)$ to each object X and a morphism $F(f) : F(X) \to F(Y)$ to each $f : X \to Y$, such that*

$$F(\mathrm{id}_X) = \mathrm{id}_{F(X)}, \qquad F(g \circ f) = F(g) \circ F(f) \quad \text{for all composable } f, g.$$

For example, forgetting the A-action gives a functor $\operatorname{Mod} A \to \operatorname{Mod} \Bbbk$: it keeps the underlying vector spaces and linear maps. The identity functor $\mathrm{id}_{\mathcal{C}}$ leaves every object and morphism unchanged; functors compose by applying one after the other. Every functor preserves isomorphisms, since $F(f^{-1})$ is inverse to $F(f)$.

The opposite category $\mathcal{C}^{\mathrm{op}}$ has the same objects as $\mathcal{C}$, with $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X,Y) = \mathrm{Hom}_{\mathcal{C}}(Y,X)$ and composition reversed. A contravariant functor from $\mathcal{C}$ to $\mathcal{D}$ means a functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus it sends $f : X \to Y$ to $F(f) : F(Y) \to F(X)$ and satisfies $F(g \circ f) = F(f) \circ F(g)$. Duals will provide our main examples.

Definition 1.6. *For functors $F, G : \mathcal{C} \to \mathcal{D}$, a natural transformation $\eta : F \to G$ is a family of morphisms $\eta_X : F(X) \to G(X)$ such that*

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ {\scriptstyle F(f)}\downarrow & & \downarrow{\scriptstyle G(f)} \\ F(Y) & \xrightarrow[\eta_Y]{} & G(Y) \end{array} \qquad G(f) \circ \eta_X = \eta_Y \circ F(f) \quad \text{for every } f : X \to Y. \tag{1.1}$$

If every η_X is an isomorphism, η is a natural isomorphism, written $F \cong G$.

The square in (1.1) is said to *commute*: its two routes from $F(X)$ to $G(Y)$ give the same morphism. For example, for fixed $c \in \Bbbk$, the maps $c\,\mathrm{id}_M : M \to M$ define a natural transformation $\mathrm{id}_{\operatorname{Mod} A} \to \mathrm{id}_{\operatorname{Mod} A}$ because $f(cm) = cf(m)$ for every module homomorphism $f : M \to N$. It is a natural isomorphism when $c \neq 0$.

A functor $F : \mathcal{C} \to \mathcal{D}$ is an *equivalence* if there is a functor $G : \mathcal{D} \to \mathcal{C}$ with natural isomorphisms $GF \cong \mathrm{id}_{\mathcal{C}}$ and $FG \cong \mathrm{id}_{\mathcal{D}}$. A *duality* between $\mathcal{C}$ and $\mathcal{D}$ is an equivalence $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Thus an equivalence recovers objects and maps up to natural isomorphism; a duality also reverses the direction of maps.

Definition 1.7. *Let A be a $\Bbbk$-algebra, and let $L, M, N \in \operatorname{Mod} A$, with $i : L \to M$ and $p : M \to N$ module homomorphisms. The sequence $L \xrightarrow{i} M \xrightarrow{p} N$ is exact at M if*

$$\operatorname{Im} i = \operatorname{Ker} p, \qquad \textit{that is,} \qquad \{i(\ell) \mid \ell \in L\} = \{m \in M \mid p(m) = 0\}.$$

A sequence of module homomorphisms is exact if it is exact at every term with an incoming and an outgoing map. A short exact sequence is an exact sequence

$$0 \longrightarrow L \xrightarrow{i} M \xrightarrow{p} N \longrightarrow 0, \qquad \operatorname{Ker} i = 0, \quad \operatorname{Im} i = \operatorname{Ker} p, \quad \operatorname{Im} p = N. \tag{1.2}$$

In (1.2), the map i identifies L with the submodule $\operatorname{Ker} p \subset M$, and p gives $M/i(L) \cong N$. For every submodule $L \subset M$, the inclusion and quotient maps give the short exact sequence $0 \to L \to M \to M/L \to 0$.

Definition 1.8. *Let A, B be $\Bbbk$-algebras. A contravariant functor $F : (\operatorname{Mod} A)^{\mathrm{op}} \to \operatorname{Mod} B$ is exact if $F(0) = 0$ and it sends every short exact sequence (1.2) to a short exact sequence*

$$0 \longrightarrow F(N) \xrightarrow{F(p)} F(M) \xrightarrow{F(i)} F(L) \longrightarrow 0.$$

An exact contravariant functor exchanges injective and surjective maps:

$$f \text{ injective} \implies F(f) \text{ surjective}, \qquad f \text{ surjective} \implies F(f) \text{ injective} \quad (f : M \to N).$$

Indeed, apply F to $0 \to M \xrightarrow{f} N \to N/f(M) \to 0$ in the first case, and to $0 \to \operatorname{Ker} f \to M \xrightarrow{f} N \to 0$ in the second. Contravariance alone reverses arrows; without exactness, the two implications need not hold.

Example 1.9 ($\Bbbk$-linear duality). *For a vector space V, define its $\Bbbk$-linear dual by $D(V) := \operatorname{Hom}_{\Bbbk}(V, \Bbbk)$. For a linear map $u : V \to W$, set*

$$D(u) : D(W) \longrightarrow D(V), \qquad \lambda \longmapsto \lambda \circ u.$$

The identities $D(\mathrm{id}_V) = \mathrm{id}_{D(V)}$ and $D(v \circ u) = D(u) \circ D(v)$ make $D : (\operatorname{Mod} \Bbbk)^{\mathrm{op}} \to \operatorname{Mod} \Bbbk$ a functor. The functor D is exact. For a short exact sequence (1.2) of vector spaces, $D(p)$ is injective because p is surjective, and $D(i)$ is surjective because a functional on a subspace extends to the whole vector space. If $\lambda \in D(M)$ satisfies $\lambda \circ i = 0$, then

$$\mu : N \longrightarrow \Bbbk, \qquad p(m) \longmapsto \lambda(m)$$

is well-defined since $\operatorname{Ker} p = i(L)$, and $\lambda = \mu \circ p = D(p)(\mu)$. Conversely, $p \circ i = 0$ gives $D(i) \circ D(p) = 0$, so $\operatorname{Ker} D(i) = \operatorname{Im} D(p)$. Thus D exchanges injective and surjective linear maps.

On finite-dimensional spaces, the canonical map to the bidual gives a natural isomorphism, which we denote by coev*:*

$$\mathrm{coev} : \mathrm{id}_{\operatorname{mod} \Bbbk} \xrightarrow{\sim} D \circ D, \qquad \mathrm{coev}_V : V \longrightarrow D(D(V)), \quad x \longmapsto (\lambda \mapsto \lambda(x)).$$

Indeed, if $x \neq 0$, some $\lambda \in D(V)$ satisfies $\lambda(x) \neq 0$, so coev_V is injective; equal finite dimensions give surjectivity. Naturality follows from

$$D(D(u))(\mathrm{coev}_V(x))(\lambda) = \lambda(u(x)) = \mathrm{coev}_W(u(x))(\lambda) \quad (x \in V,\ \lambda \in D(W)).$$

We write $\mathrm{ev}_V := \mathrm{coev}_V^{-1} : D(D(V)) \to V$ for the inverse: $\mathrm{ev}_V(\Phi)$ is the unique $x \in V$ such that $\Phi(\lambda) = \lambda(x)$ for every $\lambda \in D(V)$. Consequently, $D : (\mathrm{mod}\,\Bbbk)^{\mathrm{op}} \to \mathrm{mod}\,\Bbbk$ is a duality.

1.3 Semisimple algebras

Definition 1.10. *A module L is simple if $L \neq 0$ and its only submodules are 0 and L. A module is semisimple if it is a direct sum of simple modules. An algebra B is semisimple if it is semisimple as a right module B_B over itself. The Jacobson radical $\mathrm{rad}\,B$ is the intersection of all maximal right ideals of B.*

We use the following standard ring-theoretic facts without proof. Recall that a *division $\Bbbk$-algebra* is a non-zero $\Bbbk$-algebra whose non-zero elements are invertible.

Theorem 1.11 (Artin–Wedderburn). *For a non-zero finite-dimensional $\Bbbk$-algebra B,*

$$B \text{ is semisimple} \iff \mathrm{rad}\,B = 0 \iff B \cong \prod_{i=1}^{r} \mathrm{Mat}_{n_i}(D_i),$$

where $r, n_i \geq 1$ and each D_i is a finite-dimensional division $\Bbbk$-algebra. If $\Bbbk$ is algebraically closed, then $D_i = \Bbbk$ for every i. For semisimple B, every left or right B-module is semisimple, and every submodule has a complement. In particular, the definition is independent of the chosen side.

Corollary 1.12. *Every finite-dimensional left or right module over a semisimple algebra S is a direct summand of $S^{\oplus n}$ for some n.*

Proof A finite generating set gives a surjection $S^{\oplus n} \twoheadrightarrow M$. By Theorem 1.11, its kernel has a complement, which maps isomorphically onto M. □

Example 1.13. *The algebras $\Bbbk^{\oplus r}$ and $\mathrm{Mat}_n(\Bbbk)$ are semisimple by Theorem 1.11. For $A = \Bbbk A_2$, the ideal $J = \Bbbk e_{12}$ satisfies $J^2 = 0$. If L is simple, then LJ is either 0 or L; the latter would give $L = LJ^2 = 0$. Hence every maximal right ideal contains J. Since $A/J \cong \Bbbk \oplus \Bbbk$, those ideals are $\Bbbk e_{11} \oplus \Bbbk e_{12}$ and $\Bbbk e_{12} \oplus \Bbbk e_{22}$. Their intersection is J, so $\mathrm{rad}\,A = \Bbbk e_{12} \neq 0$.*

Remark 1.14. An element e of an algebra A is *idempotent* if $e^2 = e$. Think of e as a local unit: e is the identity of $eAe = \{eae \mid a \in A\}$, though not necessarily of A. For a finite product $A = \prod_{i=1}^{r} A_i$, the *coordinate idempotent* $e_i = (0, \ldots, 1_{A_i}, \ldots, 0)$ acts as the identity on the ith factor and as zero on the others. They sum to 1_A and are *pairwise orthogonal*, meaning $e_i e_j = 0$ for $i \neq j$.

1.4 Graded algebras

Definition 1.15. *Let G be a group with identity e. A G-graded $\Bbbk$-algebra is a $\Bbbk$-algebra A with $1_A \in A_e$ and a decomposition*

$$A = \bigoplus_{g \in G} A_g \quad \text{such that} \quad A_g A_h \subset A_{gh} \quad \text{for all } g, h \in G.$$

An element of A_g is homogeneous of degree g.

Convention. *Unless stated otherwise, a graded algebra is $\mathbb{Z}$-graded and concentrated in non-negative degrees.*

Note that the decomposition is into a direct sum, so every element of a graded algebra A can always be written as a *finite* sum of homogeneous elements.

Example 1.16. *For the upper triangular matrix ring $\Bbbk A_n$ of Example 1.2, the decomposition*

$$(\Bbbk A_n)_d = \operatorname{span}_\Bbbk\{e_{ij} \mid j - i = d\}$$

makes $\Bbbk A_n$ a graded algebra.

Example 1.17. *The algebra $\Bbbk A_3$ also has a $\mathbb{Z}^{\oplus 2}$-grading with*

$$\deg e_{ii} = (0,0), \qquad \deg e_{12} = (1,0), \qquad \deg e_{23} = (0,1), \qquad \deg e_{13} = (1,1).$$

The homomorphism $\mathbb{Z}^{\oplus 2} \to \mathbb{Z}$, $(p,q) \mapsto p+q$, gives the grading of Example 1.16. Alternatively, setting $\deg e_{ij} = 2(j-i)$ gives a $\mathbb{Z}$-grading with $A_1 = 0$ but $A_2 \neq 0$.

Definition 1.18. *A homomorphism of graded algebras $f : A \to B$ is a $\Bbbk$-algebra homomorphism with $f(A_i) \subset B_i$ for all $i \in \mathbb{Z}$.*

Definition 1.19. *A graded algebra A is locally finite if $\dim_\Bbbk A_i < \infty$ for every $i \in \mathbb{Z}$.*

1.5 Graded modules

Definition 1.20. *Let $A = \bigoplus_{g\in G} A_g$ be a G-graded algebra. A G-graded right A-module is a right A-module M with a direct sum decomposition*

$$M = \bigoplus_{g\in G} M_g \quad \text{such that} \quad M_g A_h \subset M_{gh} \quad \text{for all } g, h \in G.$$

An element of M_g is called homogeneous of degree g.

A G-graded A-module $M = \bigoplus_{g\in G} M_g$ is locally finite if $\dim_\Bbbk M_g < \infty$ for all $g \in G$.

Unless specified otherwise, graded modules are $\mathbb{Z}$-graded; their degrees need not be non-negative.

The opposite algebra A^{op} has the same vector space as A, with multiplication $a^{\mathrm{op}} b^{\mathrm{op}} = (ba)^{\mathrm{op}}$. A left A-module is a right A^{op}-module by $m \cdot a^{\mathrm{op}} = am$. For a $\mathbb{Z}$-graded algebra, $(A^{\mathrm{op}})_i = A_i$. Thus $\operatorname{Mod}(A^{\mathrm{op}})$ is the category of left A-modules and left A-linear maps. It is not the opposite category $(\operatorname{Mod} A)^{\mathrm{op}}$: the latter keeps right modules as objects and reverses the morphisms.

Definition 1.21. *An A-bimodule is a vector space M with $\Bbbk$-bilinear left and right A-module actions satisfying $(am)b = a(mb)$ for all $a, b \in A$ and $m \in M$. It is graded if $M = \bigoplus_{j\in\mathbb{Z}} M_j$ and $A_i M_j A_k \subset M_{i+j+k}$ for all $i, j, k \in \mathbb{Z}$.*

The algebra A is a graded A-bimodule by multiplication, and each A_i is an A_0-bimodule. Bilinearity forces the left and right scalar actions to agree: $(c1_A)m = cm = m(c1_A)$ for $c \in \Bbbk$. More precisely, we should have speak of '$\Bbbk$-central A-bimodule' instead; but we will always work over a field $\Bbbk$ and so will not make such distinction.

For an A-bimodule M and idempotents $e, f \in A$, the local-unit interpretation in Remark 1.14 becomes $em = m = mf$ for $m \in eMf = \{enf \mid n \in M\}$. The left and right local units may differ: $e_{11}e_{12} = e_{12} = e_{12}e_{22}$ in $\Bbbk A_2$.

Example 1.22. *The decomposition $A = \bigoplus_{i\in\mathbb{Z}} A_i$ makes the right regular module A_A a graded module, and ${}_A A_A$ a graded bimodule.*

Definition 1.23. *Let M be a graded A-module and let $d \in \mathbb{Z}$. The (degree/grading) shift $M(d)$ is the graded A-module whose underlying A-module is M, but whose grading is*

$$M(d)_j = M_{j+d}.$$

By Definition 1.23, $1 \in A_0 = A(-d)_d$, so $A(-d)$ has a generator in degree d. If M is concentrated in degree 0, then $M(d)$ is concentrated in degree $-d$.

Example 1.24. *For $A = \Bbbk A_3$ with the grading of Example 1.16, the module $P_1 := e_{11}A$ consists of matrices supported on the first row. Its basis e_{11}, e_{12}, e_{13} has degrees $0, 1, 2$, respectively; in $P_1(-2)$, the same basis has degrees $2, 3, 4$.*

Definition 1.25. *Let M, N be graded A-modules. A homomorphism $f : M \to N$ is homogeneous of degree d if $f(M_j) \subset N_{j+d}$ for all $j \in \mathbb{Z}$. A homomorphism of graded modules means a homogeneous A-module homomorphism of degree 0, also said to preserve degrees.*

Example 1.26. *Let $S = \Bbbk e_1 \oplus \Bbbk e_2$ with componentwise multiplication, concentrated in degree 0. Write*

$$M = \bigoplus_{i \geq 0} (e_1 S)(-i) = \bigoplus_{i \geq 0} \Bbbk m_i, \qquad \deg m_i = i, \quad m_i e_1 = m_i, \quad m_i e_2 = 0,$$

and let $N = e_1 S$ be concentrated in degree 0. Define the S-linear map $f : M \to N$ by

$$f\Big(\sum_{i \geq 0} c_i m_i\Big) = \sum_{c_i \neq 0} c_i e_1 \quad \text{for finitely supported } (c_i)_{i \geq 0}$$

so that $f(m_i) = e_1 \neq 0$ for every $i \geq 0$. On the summand $(e_1 S)(-i)$, the restriction $f|_{(e_1 S)(-i)} : (e_1 S)(-i) \to e_1 S$ is the underlying identity map, of degree $-i$. A homogeneous map $M \to N$ of degree d vanishes on m_i unless $d = -i$. Hence f is not a finite sum of homogeneous maps.

Example 1.27. *For $A = \Bbbk A_3$, left multiplication by e_{12} yields a map*

$$\begin{aligned} e_{22}A &\longrightarrow e_{11}A, \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & u & v \\ 0 & 0 & 0 \end{pmatrix} &\longmapsto \begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \qquad (u, v \in \Bbbk). \end{aligned}$$

The u and v entries have degrees $0, 1$ in the source and $1, 2$ in the target, and so the map has degree 1.

Notation. *For graded right A-modules M, N, the space of homomorphisms preserving degrees is*

$$\mathrm{Hom}_A^{\mathbb{Z}}(M, N) = \{f \in \mathrm{Hom}_A(M, N) \mid f(M_j) \subset N_j \text{ for all } j \in \mathbb{Z}\}. \tag{1.3}$$

Thus $\mathrm{Hom}_A^{\mathbb{Z}}(M, N(d))$ consists of homomorphisms $M \to N$ of degree d, since $N(d)_j = N_{j+d}$.

We write

$$\begin{aligned} \mathrm{GrMod}\, A &:= \text{category of graded right } A\text{-modules with morphism spaces } \mathrm{Hom}_A^{\mathbb{Z}}(M, N), \\ \mathrm{grmod}\, A &:= \text{full subcategory of } \mathrm{GrMod}\, A \text{ consisting of locally finite modules.} \end{aligned}$$

Forgetting the grading gives a functor $\mathrm{GrMod}\, A \to \mathrm{Mod}\, A$. The distinction between its source and target is the grading on objects and the degree condition on morphisms.

For $d \in \mathbb{Z}$, the shift of Definition 1.23 is a functor

$$(d) : \mathrm{GrMod}\, A \to \mathrm{GrMod}\, A.$$

It sends $f : M \to N$ to the same underlying map $f(d) : M(d) \to N(d)$. Since $M(d)(-d) = M$ and $f(d)(-d) = f$, the shift is an equivalence with inverse $(-d)$; it restricts to an equivalence $\mathrm{grmod}\, A \to \mathrm{grmod}\, A$.

Remark 1.28. A sequence in $\mathrm{GrMod}\, A$ has maps of degree 0 and is exact if the underlying sequence in $\mathrm{Mod}\, A$ is exact. For $L, M, N \in \mathrm{GrMod}\, A$ with maps $i : L \to M$ and $p : M \to N$, the sequence $0 \to L \xrightarrow{i} M \xrightarrow{p} N \to 0$ is exact if and only if

$$0 \longrightarrow L_j \xrightarrow{i|_{L_j}} M_j \xrightarrow{p|_{M_j}} N_j \longrightarrow 0 \quad \text{is exact for every } j \in \mathbb{Z}.$$

Indeed, i and p preserve the direct sum decompositions, so their kernels and images decompose into the kernels and images of the restrictions.

Example 1.29. *Let $A = \Bbbk A_2$. The inclusion and quotient maps give a short exact sequence in* $\mathrm{GrMod}\, A$*:*

$$0 \longrightarrow \Bbbk e_{12} \xrightarrow{i} e_{11}A \xrightarrow{p} e_{11}A/\Bbbk e_{12} \longrightarrow 0, \qquad i(ce_{12}) = ce_{12}, \quad p(m) = \overline{m}.$$

The source is concentrated in degree 1*, the quotient in degree* 0*, and* $\operatorname{Ker} p = \Bbbk e_{12} = \operatorname{Im} i$*.*

Lemma 1.30. *For graded right A-modules M, N, the following inclusion is an equality when M is finitely generated:*

$$\bigoplus_{d \in \mathbb{Z}} \mathrm{Hom}_A^{\mathbb{Z}}(M, N(d)) \subset \mathrm{Hom}_A(M, N).$$

Proof For $f \in \mathrm{Hom}_A(M, N)$, define $f_d(m) = f(m)_{j+d}$ when $m \in M_j$, and extend linearly. For $a \in A_i$, we have $f_d(ma) = f(ma)_{j+i+d} = f(m)_{j+d}a$, so f_d is A-linear of degree d. The sum of the homogeneous Hom spaces is direct: on M_j, distinct degrees have values in distinct summands of N. If M is finitely generated, the homogeneous components of a finite generating set form a finite homogeneous generating set. Only finitely many f_d are non-zero on these homogeneous generators, hence on M, giving $f = \sum_d f_d$. □

1.6 Homogeneous ideals and quotients

Definition 1.31. *Let $A = \bigoplus_{i \in \mathbb{Z}} A_i$ be a graded algebra. A right ideal $I \subset A$ is* homogeneous *if*

$$I = \bigoplus_{i \in \mathbb{Z}} (I \cap A_i).$$

Equivalently, whenever $a \in I$ and $a = a_0 + \cdots + a_N$ is its decomposition into homogeneous components, each a_i also belongs to I.

For quotient algebras we need two-sided ideals. If I is a homogeneous two-sided ideal, then

$$A/I = \bigoplus_{i \in \mathbb{Z}} A_i/(I \cap A_i)$$

is again a graded algebra.

Example 1.32. *In $A = \Bbbk A_3$ with the grading of Example 1.16, the two-sided ideal $I = (e_{13}) = \Bbbk e_{13}$ is homogeneous. Writing $\overline{e}_{ij}$ for the image of e_{ij},*

$$(A/I)_0 \cong \Bbbk^{\oplus 3}, \qquad (A/I)_1 = \Bbbk \overline{e}_{12} \oplus \Bbbk \overline{e}_{23}, \qquad (A/I)_d = 0 \quad (d \geq 2).$$

In particular, $\overline{e}_{12}\overline{e}_{23} = 0$.

Example 1.33. *Let $A = \Bbbk[x]$ with $\deg x = 1$, and let $I = (x - x^2)$. The ideal I is two-sided because A is commutative. Every element of I vanishes at $x = 1$, whereas neither x nor x^2 does. Thus $x - x^2 \in I$ but $x, x^2 \notin I$, so I is not homogeneous. Writing $\overline{f} = f + I$, we have $0 \neq \overline{x} = \overline{x^2}$. The images of A_1 and A_2 therefore intersect non-trivially, so A/I does not inherit the quotient grading. As an ungraded algebra, the quotient is semisimple:*

$$\Bbbk[x]/(x - x^2) \xrightarrow{\sim} \Bbbk \oplus \Bbbk, \qquad \overline{f} \longmapsto (f(0), f(1)).$$

The inverse sends (a, b) to $a + (b - a)\overline{x}$.

Definition 1.34. *The positive degree ideal of a graded algebra A is $A_{>0} := \bigoplus_{i>0} A_i$.*[1]

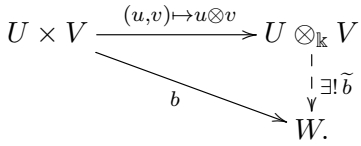
The subspace $A_{>0}$ is a homogeneous two-sided ideal. The quotient map

$$\pi : A \longrightarrow A/A_{>0} \cong A_0, \qquad \sum_{i\in\mathbb{Z}} a_i \mapsto a_0,$$

makes A_0 a graded A-bimodule concentrated in degree 0, with $a \cdot s \cdot b = \pi(a)s\pi(b)$ for $a, b \in A$ and $s \in A_0$.

1.7 Tensor products of vector spaces

Definition 1.35. *For vector spaces U, V, the tensor product $U \otimes_{\Bbbk} V$ is equipped with a bilinear map $(u, v) \mapsto u \otimes v$ such that every bilinear map $b : U \times V \to W$ has a unique linear factorisation:*



Choose bases $(u_i)_{i\in I}$ of U and $(v_j)_{j\in J}$ of V. Construct $U \otimes_{\Bbbk} V$ as the vector space with basis $\{u_i \otimes v_j \mid i \in I,\ j \in J\}$, and extend $(u_i, v_j) \mapsto u_i \otimes v_j$ bilinearly. For $b : U \times V \to W$ bilinear, the unique required map is

$$\widetilde{b} : U \otimes_{\Bbbk} V \longrightarrow W, \qquad u_i \otimes v_j \longmapsto b(u_i, v_j).$$

Let T, T' be vector spaces with bilinear maps $t : U \times V \to T$ and $t' : U \times V \to T'$ satisfying the universal property in Definition 1.35. The universal properties give unique linear maps $a : T \to T'$ and $b : T' \to T$ with $a \circ t = t'$ and $b \circ t' = t$. The universal property of (T, t), applied to the bilinear map $t : U \times V \to T$, gives a unique linear map $h : T \to T$ with $h \circ t = t$. Both $b \circ a$ and id_T have this property, since

$$(b \circ a) \circ t = b \circ t' = t = \mathrm{id}_T \circ t.$$

Hence $b \circ a = \mathrm{id}_T$. Likewise, $(a \circ b) \circ t' = a \circ t = t' = \mathrm{id}_{T'} \circ t'$, so uniqueness for (T', t') gives $a \circ b = \mathrm{id}_{T'}$. Thus the tensor product is independent of the chosen bases up to the unique isomorphism carrying $t(u, v)$ to $t'(u, v)$.

For finite bases with $|I| = r$ and $|J| = s$, we obtain $\dim_{\Bbbk}(U \otimes_{\Bbbk} V) = rs$. We use the canonical identifications

$$(U \otimes_{\Bbbk} V) \otimes_{\Bbbk} W \xrightarrow{\sim} U \otimes_{\Bbbk} (V \otimes_{\Bbbk} W), \quad (u \otimes v) \otimes w \longmapsto u \otimes (v \otimes w),$$
$$\Bbbk \otimes_{\Bbbk} U \xrightarrow{\sim} U, \qquad\qquad c \otimes u \longmapsto cu,$$

and set $U^{\otimes_{\Bbbk} 0} := \Bbbk$.

1.8 Direct sums and direct products

For a family $(M_i)_{i\in I}$ of right A-modules, the *direct product* and *direct sum* are

$$\prod_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i \text{ for all } i \in I\},$$
$$\bigoplus_{i\in I} M_i = \{(m_i)_{i\in I} \in \prod_{i\in I} M_i \mid m_i = 0 \text{ for all but finitely many } i\}.$$

[1] It is also called the augmentation ideal; we avoid that terminology.

Addition and the A-action are componentwise; the same definitions apply to vector spaces by taking $A = \Bbbk$. Thus $\bigoplus_i M_i \subset \prod_i M_i$, with equality for finite I. For example, $(1, 1, \ldots) \in \prod_{i\geq 0} \Bbbk$ does not belong to $\bigoplus_{i\geq 0} \Bbbk$. In a direct sum, an element is a finite sum of its components; no notion of convergence is involved.

Tensor products distribute over direct sums: for vector spaces U_i $(i \in I)$ and V, there is a canonical linear isomorphism

$$\Big(\bigoplus_i U_i\Big) \otimes_\Bbbk V \xrightarrow{\sim} \bigoplus_i (U_i \otimes_\Bbbk V), \qquad (u_i)_i \otimes v \longmapsto (u_i \otimes v)_i.$$

The map is defined by the universal property in Definition 1.35. If $\iota_i : U_i \to \bigoplus_j U_j$ is the inclusion, the inverse sends $u \otimes v$ in summand i to $\iota_i(u) \otimes v$. The same construction applies in the other tensor variable.

For direct products, the corresponding map need not be an isomorphism.

Lemma 1.36. *For vector spaces U_i $(i \in I)$ and V, the coordinate projections define a linear map*

$$\Big(\prod_i U_i\Big) \otimes_\Bbbk V \longrightarrow \prod_i (U_i \otimes_\Bbbk V), \qquad (u_i)_i \otimes v \longmapsto (u_i \otimes v)_i. \tag{1.4}$$

This map is injective, and is an isomorphism if I is finite or V is finite-dimensional. The same statements hold with the two tensor factors reversed.

Proof The map is defined by the universal property in Definition 1.35. Every tensor can be written $\sum_{\ell=1}^{r} u^{(\ell)} \otimes v_\ell$ with the v_ℓ linearly independent: choose a basis of the span of its finitely many second factors. If its image is zero, then $\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell = 0$ for every i. The tensor-product basis construction gives $u_i^{(\ell)} = 0$ for all i, ℓ, proving injectivity. If $v_1, \ldots, v_r$ is a basis of V, the inverse is

$$\prod_i (U_i \otimes_\Bbbk V) \longrightarrow \Big(\prod_i U_i\Big) \otimes_\Bbbk V, \qquad \Big(\sum_{\ell=1}^{r} u_i^{(\ell)} \otimes v_\ell\Big)_i \longmapsto \sum_{\ell=1}^{r} (u_i^{(\ell)})_i \otimes v_\ell.$$

For finite I, products equal sums. Reversing the tensor factors gives the other statements, since $U \otimes_\Bbbk V \to V \otimes_\Bbbk U$, $u \otimes v \mapsto v \otimes u$, is an isomorphism. □

Example 1.37. *The comparison in* (1.4) *need not be surjective when V is infinite-dimensional. Take $U_i = \Bbbk$ for $i \geq 1$ and $V = \bigoplus_{j\geq 1} \Bbbk v_j$. Under $\Bbbk \otimes_\Bbbk V \cong V$, the image of the product comparison in* (1.4) *is*

$$\left\{ (w_i)_i \in \prod_{i\geq 1} V \;\middle|\; \dim_\Bbbk \operatorname{span}_\Bbbk\{w_i \mid i \geq 1\} < \infty \right\}.$$

Indeed, the image of $\sum_{\ell=1}^{r} (c_i^{(\ell)})_i \otimes w_\ell$ has every entry in $\operatorname{span}_\Bbbk\{w_1, \ldots, w_r\}$. Conversely, expand all entries in a basis of their common finite-dimensional span. Since the v_i are linearly independent, $(v_i)_i$ is not in the image.

1.9 Enveloping algebras and bimodules

Definition 1.38. *For a $\Bbbk$-algebra A, its enveloping algebra is $A^{\mathrm{en}} := A \otimes_\Bbbk A^{\mathrm{op}}$, with multiplication extended bilinearly from*

$$(a \otimes b^{\mathrm{op}})(a' \otimes (b')^{\mathrm{op}}) = aa' \otimes (b'b)^{\mathrm{op}}, \qquad 1_{A^{\mathrm{en}}} = 1_A \otimes 1_A^{\mathrm{op}} \quad (a, b, a', b' \in A).$$

An A-bimodule becomes a right A^{en}-module by

$$m \cdot (a \otimes b^{\mathrm{op}}) = bma \quad (a, b \in A,\ m \in M). \tag{1.5}$$

The first tensor factor records the right action, and the second the left action:

$$\big(m \cdot (a \otimes b^{\mathrm{op}})\big) \cdot (a' \otimes (b')^{\mathrm{op}}) = b'bmaa' = m \cdot \big(aa' \otimes (b'b)^{\mathrm{op}}\big).$$

Conversely, for $M \in \operatorname{Mod} A^{\mathrm{en}}$, define $ma = m \cdot (a \otimes 1)$ and $bm = m \cdot (1 \otimes b^{\mathrm{op}})$. The two actions commute because $a \otimes 1$ and $1 \otimes b^{\mathrm{op}}$ commute in A^{en}. For A-bimodules M, N, a linear map $u : M \to N$ satisfies

$$u(bma) = bu(m)a \quad (a, b \in A,\ m \in M) \quad \Longleftrightarrow \quad u \in \operatorname{Hom}_{A^{\mathrm{en}}}(M, N).$$

Thus $\operatorname{Mod} A^{\mathrm{en}}$ is also the category of A-bimodules and maps preserving both actions.

Grant–Iyama [GI20, §2, p. 2591] identify bimodules with left modules over A^{en}, using $(a \otimes b^{\mathrm{op}})m = amb$. Our right module convention is (1.5): $m \cdot (a \otimes b^{\mathrm{op}}) = bma$. Both describe the same bimodules; the side on which A^{en} acts determines the formula.

Remark 1.39. For a finite-dimensional semisimple $\Bbbk$-algebra S, semisimplicity of S^{en} requires an additional condition. For a field extension $K/\Bbbk$, give $K \otimes_{\Bbbk} S$ the multiplication $(c \otimes s)(d \otimes t) = cd \otimes st$. The algebra S is called *separable over* $\Bbbk$ if $K \otimes_{\Bbbk} S$ is semisimple for every field extension $K/\Bbbk$. By [DK94, Theorem 6.1.2, p. 105],

$$S^{\mathrm{en}} \text{ is semisimple} \quad \Longleftrightarrow \quad S \text{ is separable over } \Bbbk.$$

For a product of matrix algebras, every field extension $K/\Bbbk$ gives

$$K \otimes_{\Bbbk} \prod_{j=1}^{r} \operatorname{Mat}_{n_j}(\Bbbk) \xrightarrow{\sim} \prod_{j=1}^{r} \operatorname{Mat}_{n_j}(K), \qquad c \otimes (a_j)_j \longmapsto (ca_j)_j.$$

The target is semisimple by Theorem 1.11. Thus products of matrix algebras over $\Bbbk$ are separable; in particular, every finite-dimensional semisimple algebra over an algebraically closed field is separable. Semisimplicity of S alone does not imply semisimplicity of S^{en} over an arbitrary field.

Example 1.40. *Let $\Bbbk = \mathbb{F}_2(t)$, where t is an indeterminate, and let $S = \Bbbk[u]/(u^2 - t)$, also writing u for its residue class. The element t is not a square in $\Bbbk$: an equality $p(t)^2 = tq(t)^2$ for non-zero polynomials would give $2\deg p = 1 + 2\deg q$. Thus $u^2 - t$ is irreducible and S is a field, hence semisimple. However, since S is commutative and has characteristic 2,*

$$S^{\mathrm{en}} = S \otimes_{\Bbbk} S \cong S[v]/(v^2 - t) = S[v]/((v - u)^2) \cong S[\varepsilon]/(\varepsilon^2).$$

Here the first tensor factor supplies the coefficients, v corresponds to $1 \otimes u$, and $\varepsilon = 1 \otimes u - u \otimes 1$. The ideal $S\varepsilon$ is non-zero, has square zero, and is the unique maximal ideal, so $\operatorname{rad}(S^{\mathrm{en}}) = S\varepsilon \neq 0$. Thus S^{en} is not semisimple, and the field extension $S/\Bbbk$ already shows that S is not separable over $\Bbbk$.

1.10 Graded $\Bbbk$-linear duals

A linear functional on a graded vector space need not have a degree: it may be non-zero on infinitely many homogeneous components. To obtain a graded vector space, we retain only finite sums of homogeneous functionals.

Definition 1.41. *For a graded vector space $V = \bigoplus_{i\in\mathbb{Z}} V_i$, its graded $\Bbbk$-linear dual is the graded vector space defined using the linear dual D of Example 1.9 by*

$$\mathbb{D}(V) = \bigoplus_{i\in\mathbb{Z}} \mathbb{D}(V)_i \qquad \text{with} \qquad \mathbb{D}(V)_i = D(V_{-i}).$$

> For general infinite-dimensional V, we have $\mathbb{D}(V) \subset D(V)$, whereas restriction to the homogeneous components gives $D(V) \cong \prod_{i\in\mathbb{Z}} D(V_i)$. An arbitrary family of functionals on the V_i defines a functional on V, since each vector has only finitely many non-zero components. These spaces coincide when V is finite-dimensional.

Linear duality reverses the side of a module action. For a right A-module M and a left A-module N, define

$$(af)(m) := f(ma), \qquad (ga)(n) := g(an) \quad (a \in A,\ f \in D(M),\ g \in D(N)). \tag{1.6}$$

Here $m \in M$ and $n \in N$; the first formula makes $D(M)$ a left module and the second makes $D(N)$ a right module. For graded modules, the same formulas preserve the graded $\Bbbk$-linear dual: if $a \in A_p$ and $f \in \mathbb{D}(M)_q$, then af vanishes outside M_{-p-q} and hence has degree $p+q$.

For a module homomorphism $u : M \to N$, precomposition defines $D(u) : D(N) \to D(M)$ by $D(u)(f) = f \circ u$. If u preserves degrees, this restricts to $\mathbb{D}(u) : \mathbb{D}(N) \to \mathbb{D}(M)$, also of degree 0. Thus we obtain functors

$$D : (\operatorname{Mod} A)^{\mathrm{op}} \longrightarrow \operatorname{Mod}(A^{\mathrm{op}}), \qquad \mathbb{D} : (\operatorname{GrMod} A)^{\mathrm{op}} \longrightarrow \operatorname{GrMod}(A^{\mathrm{op}}). \tag{1.7}$$

Indeed, $D(\mathrm{id}_M) = \mathrm{id}_{D(M)}$ and $D(v \circ u) = D(u) \circ D(v)$ for composable u, v, and the same identities hold for $\mathbb{D}$. Replacing A by A^{op} gives the corresponding functors from left modules to right modules.

Example 1.42. *Let $A = \Bbbk A_3$ and $P_1 = e_{11}A$ as in Example 1.24. The functionals $\varepsilon_{1j}(e_{1\ell}) = \delta_{j\ell}$ for $1 \le j, \ell \le 3$ form a basis of $\mathbb{D}(P_1)$, with*

$$\deg \varepsilon_{1j} = 1 - j, \qquad e_{12}\varepsilon_{12} = \varepsilon_{11}, \qquad e_{23}\varepsilon_{13} = \varepsilon_{12}, \qquad e_{jj}\varepsilon_{1j} = \varepsilon_{1j}.$$

The actions follow from (1.6).

Lemma 1.43. *The functor $\mathbb{D}$ in* (1.7) *restricts to a duality $(\operatorname{grmod} A)^{\mathrm{op}} \to \operatorname{grmod}(A^{\mathrm{op}})$. The coevaluation map and its inverse are natural isomorphisms*

$$\begin{aligned} \mathrm{coev}_M : M &\xrightarrow{\sim} \mathbb{D}(\mathbb{D}(M)), \quad m \longmapsto (f \mapsto f(m)), \\ \mathrm{ev}_M := \mathrm{coev}_M^{-1} : \mathbb{D}(\mathbb{D}(M)) &\xrightarrow{\sim} M \qquad\qquad (M \in \operatorname{grmod} A). \end{aligned} \tag{1.8}$$

Thus $\mathrm{ev}_M(\Phi)$ is the unique $m \in M$ such that $\Phi(f) = f(m)$ for every $f \in \mathbb{D}(M)$. Moreover, $\mathbb{D}$ takes every short exact sequence $0 \to L \to M \to N \to 0$ to a short exact sequence $0 \to \mathbb{D}(N) \to \mathbb{D}(M) \to \mathbb{D}(L) \to 0$.

Proof The map coev_M in (1.8) is an isomorphism in each degree by Example 1.9, and (1.6) makes it A-linear. For $u : M \to N$ of degree 0, $m \in M$, and $f \in \mathbb{D}(N)$,

$$\big(\mathbb{D}(\mathbb{D}(u))(\mathrm{coev}_M(m))\big)(f) = \mathrm{coev}_M(m)(f \circ u) = f(u(m)) = \mathrm{coev}_N(u(m))(f).$$

Hence $\mathbb{D}(\mathbb{D}(u)) \circ \mathrm{coev}_M = \mathrm{coev}_N \circ u$, which is naturality as in (1.1). Taking inverses of the coevaluation maps gives $\mathrm{ev}_N \circ \mathbb{D}(\mathbb{D}(u)) = u \circ \mathrm{ev}_M$. Replacing A by A^{op} gives the other natural isomorphism required for a duality. For a short exact sequence $0 \to L \xrightarrow{i} M \xrightarrow{p} N \to 0$ in $\operatorname{GrMod} A$, apply the exactness of D from Example 1.9 in each degree. The resulting sequences are

$$0 \longrightarrow D(N_{-d}) \xrightarrow{D(p|_{M_{-d}})} D(M_{-d}) \xrightarrow{D(i|_{L_{-d}})} D(L_{-d}) \longrightarrow 0 \quad (d \in \mathbb{Z}).$$

By Remark 1.28, these give the short exact sequence $0 \to \mathbb{D}(N) \xrightarrow{\mathbb{D}(p)} \mathbb{D}(M) \xrightarrow{\mathbb{D}(i)} \mathbb{D}(L) \to 0$. □

The same argument without grading gives a duality between finite-dimensional right and left A-modules. If A is finite-dimensional, these are precisely the finitely generated modules, so $D : (\operatorname{mod} A)^{\mathrm{op}} \to \operatorname{mod}(A^{\mathrm{op}})$ is a duality. Without finite dimensionality for D, or local finiteness for $\mathbb{D}$, coevaluation need not be surjective, so its inverse ev need not exist.

1.11 Left, right, and two-sided duals over S

Fix a finite-dimensional semisimple $\Bbbk$-algebra S, which is regarded as concentrated in degree 0 when treated as a graded algebra. Besides the $\Bbbk$-linear dual, we will also work with duals defined by left S-linear maps and right S-linear maps. We first define these duals without grading, then apply the constructions in each degree.

Definition 1.44. *For a right S-module M and a left S-module N, define their S-duals by*

$$\begin{aligned} M^* &:= \operatorname{Hom}_S(M, S), \qquad (sf)(m) := sf(m), \\ {}^*N &:= \operatorname{Hom}_{S^{\mathrm{op}}}(N, S), \qquad (gs)(n) := g(n)s. \end{aligned}$$

Here $s \in S$, $m \in M$, $n \in N$, $f \in M^*$, and $g \in {}^*N$.

$$M \in \operatorname{Mod} S \implies M^* \in \operatorname{Mod}(S^{\mathrm{op}}), \qquad N \in \operatorname{Mod}(S^{\mathrm{op}}) \implies {}^*N \in \operatorname{Mod} S.$$

The same implications hold with Mod replaced by mod, since S is finite-dimensional.

For a right module map $u : M \to N$, define $u^* : N^* \to M^*$ by $u^*(f) = f \circ u$; for a left module map $u : M \to N$, the same recipe defines ${}^*u : {}^*N \to {}^*M$. Precomposition gives functors

$$(-)^* : (\operatorname{Mod} S)^{\mathrm{op}} \longrightarrow \operatorname{Mod}(S^{\mathrm{op}}), \qquad {}^*(-) : (\operatorname{Mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{Mod} S.$$

The grading is introduced in the same way as in Definition 1.41.

Definition 1.45. *For graded right and left S-modules M, respectively, define their graded S-duals by*

$$M^{(*)} := \bigoplus_{i \in \mathbb{Z}} (M_{-i})^*, \qquad {}^{(*)}M := \bigoplus_{i \in \mathbb{Z}} {}^*(M_{-i}),$$

where each indicated summand has degree i.

For a degree 0 right S-module map $u : M \to N$, define $u^{(*)} : N^{(*)} \to M^{(*)}$ by

$$(u^{(*)})_i : (N_{-i})^* \longrightarrow (M_{-i})^*, \qquad f \longmapsto f \circ u|_{M_{-i}} \quad (i \in \mathbb{Z}).$$

For a degree 0 left module map $u : M \to N$, the same formula defines ${}^{(*)}u : {}^{(*)}N \to {}^{(*)}M$. With the actions from Definition 1.44, we obtain functors

$$\begin{aligned} (-)^{(*)} &: (\operatorname{GrMod} S)^{\mathrm{op}} \longrightarrow \operatorname{GrMod}(S^{\mathrm{op}}), \\ {}^{(*)}(-) &: (\operatorname{GrMod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{GrMod} S. \end{aligned} \tag{1.9}$$

Each functor in (1.9) restricts to locally finite modules: replace GrMod by grmod in its source and target. Indeed, for $R = S$ or S^{op} and $M \in \operatorname{grmod} R$,

$$\dim_{\Bbbk} \operatorname{Hom}_R(M_{-i}, R) \le (\dim_{\Bbbk} M_{-i})(\dim_{\Bbbk} R) < \infty \quad (i \in \mathbb{Z}). \tag{1.10}$$

Parentheses on the stars distinguish the graded constructions from the full S-duals. In the notation (1.3),

$$M^{(*)} = \bigoplus_{i \in \mathbb{Z}} \operatorname{Hom}_S^{\mathbb{Z}}(M, S(i)), \qquad {}^{(*)}M = \bigoplus_{i \in \mathbb{Z}} \operatorname{Hom}_{S^{\mathrm{op}}}^{\mathbb{Z}}(M, S(i)).$$

Lemma 1.46. *Coevaluation gives natural isomorphisms of functors*

$$\mathrm{id}_{\mathrm{mod}\, S} \xrightarrow{\sim} {}^*((-)^*), \qquad \mathrm{id}_{\mathrm{mod}(S^{\mathrm{op}})} \xrightarrow{\sim} ({}^*(-))^*,$$

with components

$$\mathrm{coev}_M : M \xrightarrow{\sim} {}^*(M^*), \quad m \longmapsto (f \mapsto f(m)) \quad (M \in \mathrm{mod}\, S),$$
$$\mathrm{coev}_N : N \xrightarrow{\sim} ({}^*N)^*, \quad n \longmapsto (g \mapsto g(n)) \quad (N \in \mathrm{mod}(S^{\mathrm{op}})).$$

Thus $(-)^* : (\mathrm{mod}\, S)^{\mathrm{op}} \to \mathrm{mod}(S^{\mathrm{op}})$ *and* ${}^*(-) : (\mathrm{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \to \mathrm{mod}\, S$ *are inverse dualities.*

Proof For $u : M \to N$ in $\mathrm{mod}\, S$, $m \in M$, and $f \in N^*$,

$$\big({}^*(u^*)(\mathrm{coev}_M(m))\big)(f) = (u^* f)(m) = f(u(m)) = \mathrm{coev}_N(u(m))(f).$$

Hence ${}^*(u^*) \circ \mathrm{coev}_M = \mathrm{coev}_N \circ u$. For $M = S$, coevaluation is an isomorphism: its inverse sends $F \in {}^*(S^*)$ to $F(\mathrm{id}_S)$. Indeed, every $f \in S^*$ satisfies $f(s) = f(1)s$, so $F(f) = f(1)F(\mathrm{id}_S) = f(F(\mathrm{id}_S))$. Restriction to summands identifies the dual of a finite direct sum with the direct sum of the duals; coevaluation therefore is an isomorphism for $S^{\oplus n}$. By Corollary 1.12, $M \oplus L \cong S^{\oplus n}$ for some L. Under the direct sum identifications, $\mathrm{coev}_{M\oplus L} = \mathrm{coev}_M \oplus \mathrm{coev}_L$, so coev_M is an isomorphism. Replacing S by S^{op} proves the left module statement. □

Applying Lemma 1.46 in each degree gives natural isomorphisms

$$\mathrm{id}_{\mathrm{grmod}\, S} \xrightarrow{\sim} {}^{(*)}((-)^{(*)}), \qquad \mathrm{id}_{\mathrm{grmod}(S^{\mathrm{op}})} \xrightarrow{\sim} ({}^{(*)}(-))^{(*)}.$$

Their components are again given by $m \mapsto (f \mapsto f(m))$. Thus the functors in (1.9) restrict to inverse dualities on locally finite graded modules.

Recall from Definition 1.38 and (1.5) that

$$S^{\mathrm{en}} = S \otimes_{\Bbbk} S^{\mathrm{op}}, \qquad m \cdot (s \otimes t^{\mathrm{op}}) = tms,$$

so S-bimodules are the objects of $\mathrm{Mod}\, S^{\mathrm{en}}$. All three spaces $D(M)$, M^*, and ${}^*M$ inherit bimodule structures:

$$(s\lambda t)(m) = \lambda(tms), \qquad (sft)(m) = sf(tm), \qquad (sgt)(m) = g(ms)t, \tag{1.11}$$

where $\lambda \in D(M)$, $f \in M^*$, $g \in {}^*M$, $s, t \in S$, and $m \in M$.

For a finite-dimensional S-bimodule M, coevaluation in Lemma 1.46 preserves both actions:

$$(s\,\mathrm{coev}_M(m)\,t)(f) = f(sm)t = f(smt) = \mathrm{coev}_M(smt)(f) \quad (s, t \in S,\ m \in M,\ f \in M^*).$$

The graded coevaluation maps preserve both actions by the same calculation in each degree.

Definition 1.47. *The two-sided dual of an S-bimodule M is*

$$M^{*\mathrm{en}} := \mathrm{Hom}_{S^{\mathrm{en}}}(M, S^{\mathrm{en}}), \qquad (sFt)(m) := (s \otimes t^{\mathrm{op}})F(m) \quad (s, t \in S,\ F \in M^{*\mathrm{en}},\ m \in M).$$

In Definition 1.47, M has the right action (1.5), so $F(tms) = F(m)(s \otimes t^{\mathrm{op}})$. The output is a left S^{en}-module by multiplication on the values, which gives the displayed bimodule action. The two-sided dual takes values in S^{en}, not in S. On bimodule morphisms it acts by precomposition, as do D, $(-)^*$, and ${}^*(-)$; each is a contravariant functor on the category of S-bimodules. The two-sided dual records both module actions in the target algebra S^{en}.

For a graded S-bimodule M, define its *graded two-sided dual* by

$$M^{(*_{\mathrm{en}})} := \bigoplus_{i\in\mathbb{Z}} (M_{-i})^{*_{\mathrm{en}}}, \qquad (M^{(*_{\mathrm{en}})})_i = (M_{-i})^{*_{\mathrm{en}}}.$$

The bimodule actions in (1.11) and Definition 1.47 give functors

$$(-)^{(*)},\ {}^{(*)}(-),\ (-)^{(*_{\mathrm{en}})} : (\mathrm{GrMod}\, S^{\mathrm{en}})^{\mathrm{op}} \longrightarrow \mathrm{GrMod}\, S^{\mathrm{en}}. \tag{1.12}$$

On a degree 0 bimodule map $u : M \to N$, each functor acts by precomposition in each degree; in particular,

$$(u^{(*_{\mathrm{en}})})_i : (N_{-i})^{*_{\mathrm{en}}} \longrightarrow (M_{-i})^{*_{\mathrm{en}}}, \qquad F \longmapsto F \circ u|_{M_{-i}}.$$

These functors restrict to locally finite bimodules: the dimension bound (1.10) also applies with $R = S^{\mathrm{en}}$.

The four duals above follow the distinction in Grant–Iyama [GI20, §2.1, pp. 2592–2593], with the following notation changes:

construction	Grant–Iyama	these notes
$\Bbbk$-linear dual	M^*	$D(M)$
left S-linear dual	M^{*_ℓ}	${}^*M$
right S-linear dual	M^{*_r}	M^*
two-sided dual	$M^\vee$	$M^{*_{\mathrm{en}}}$

The last row also changes the convention for enveloping modules. Their $F : M \to S^{\mathrm{en}}$ is left S^{en}-linear. Define

$$\sigma : S^{\mathrm{en}} \longrightarrow S^{\mathrm{en}}, \qquad a \otimes b^{\mathrm{op}} \longmapsto b \otimes a^{\mathrm{op}}.$$

Since $\sigma(xy) = \sigma(y)\sigma(x)$ for $x, y \in S^{\mathrm{en}}$, the map $\sigma \circ F : M \to S^{\mathrm{en}}$ satisfies

$$(\sigma \circ F)(tms) = (\sigma \circ F)(m)(s \otimes t^{\mathrm{op}}) \quad (s, t \in S,\ m \in M),$$

and belongs to our $M^{*_{\mathrm{en}}}$. Thus $M^\vee \to M^{*_{\mathrm{en}}}$, $F \mapsto \sigma \circ F$, translates their two-sided dual into our convention. In particular, their star denotes the linear dual; our star denotes the right S-dual.

1.12 Comparing the duals

To identify S-dual with $\Bbbk$-dual, we need a suitable linear functional $S \to \Bbbk$.

Definition 1.48. *A symmetrising form on a finite-dimensional $\Bbbk$-algebra S is a linear map $\tau : S \to \Bbbk$ satisfying*

$$\tau(st) = \tau(ts) \quad \textit{for all } s, t \in S, \qquad \big(\tau(st) = 0 \textit{ for all } t \in S\big) \implies s = 0.$$

An algebra admitting such a form is symmetric.

The second condition in Definition 1.48 says that the bilinear form $(s, t) \mapsto \tau(st)$ is *non-degenerate*: every non-zero $s \in S$ pairs non-trivially with some $t \in S$.

In practice (including within the scope of these lectures), we only really use the following explicit symmetrising forms.

Example 1.49. *The following maps are symmetrising forms:*

$$\tau : \Bbbk^{\oplus r} \longrightarrow \Bbbk, \quad (a_1, \dots, a_r) \longmapsto \sum_{j=1}^{r} a_j,$$

$$\tau : \mathrm{Mat}_n(\Bbbk) \longrightarrow \Bbbk, \qquad a \longmapsto \mathrm{Tr}(a) := \sum_{i=1}^{n} a_{ii},$$

and more generally, $\tau : \prod_{j=1}^{r} \mathrm{Mat}_{n_j}(\Bbbk) \longrightarrow \Bbbk, \quad (a_1, \dots, a_r) \longmapsto \sum_{j=1}^{r} \mathrm{Tr}(a_j).$

Theorem 1.50 (Comparison of duals). *Choose a symmetrising form $\tau : S \to \Bbbk$. The right and left S-dual functors are naturally isomorphic to the corresponding $\Bbbk$-linear dual functors, both without grading and with grading.*

Ungraded duals. *For $M \in \operatorname{mod} S$ and $N \in \operatorname{mod}(S^{\mathrm{op}})$, the maps*

$$\alpha_M : M^* \longrightarrow D(M), \quad f \longmapsto \tau \circ f, \qquad \beta_N : {}^*N \longrightarrow D(N), \quad g \longmapsto \tau \circ g \tag{1.13}$$

are S-module isomorphisms and give natural isomorphisms

$$\begin{aligned} &\alpha : (-)^* \xrightarrow{\sim} D \quad \textit{of functors } (\operatorname{mod} S)^{\mathrm{op}} \longrightarrow \operatorname{mod}(S^{\mathrm{op}}), \\ &\beta : {}^*(-) \xrightarrow{\sim} D \quad \textit{of functors } (\operatorname{mod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{mod} S. \end{aligned} \tag{1.14}$$

Graded duals. *Composition with τ in each degree gives natural isomorphisms*

$$\begin{aligned} &\alpha : (-)^{(*)} \xrightarrow{\sim} \mathbb{D} \quad \textit{of functors } (\operatorname{grmod} S)^{\mathrm{op}} \longrightarrow \operatorname{grmod}(S^{\mathrm{op}}), \\ &\beta : {}^{(*)}(-) \xrightarrow{\sim} \mathbb{D} \quad \textit{of functors } (\operatorname{grmod}(S^{\mathrm{op}}))^{\mathrm{op}} \longrightarrow \operatorname{grmod} S. \end{aligned} \tag{1.15}$$

For $M \in \operatorname{grmod} S$ and $N \in \operatorname{grmod}(S^{\mathrm{op}})$, the degree i components are $\alpha_{M_{-i}}$ and $\beta_{N_{-i}}$, respectively.

Two-sided duals. *If M is a finite-dimensional S-bimodule, α_M and β_M preserve both actions from* (1.11)*. There is also a natural isomorphism $\gamma : (-)^{*\mathrm{en}} \xrightarrow{\sim} (-)^*$ given by*

$$\gamma_M : M^{*\mathrm{en}} \to M^*, \qquad F \mapsto \big(m \mapsto \sum_j a_j \tau(b_j)\big) \quad \textit{for} \quad F(m) = \sum_j a_j \otimes b_j^{\mathrm{op}}. \tag{1.16}$$

Consequently, on finite-dimensional S-bimodules all four dual functors are naturally isomorphic:

$$(-)^{*\mathrm{en}} \xrightarrow[\gamma]{\sim} (-)^* \xrightarrow[\alpha]{\sim} D \xleftarrow[\beta]{\sim} {}^*(-).$$

Applying γ in each degree gives the corresponding natural isomorphisms on locally finite graded S-bimodules:

$$(-)^{(*\mathrm{en})} \xrightarrow[\gamma]{\sim} (-)^{(*)} \xrightarrow[\alpha]{\sim} \mathbb{D} \xleftarrow[\beta]{\sim} {}^{(*)}(-).$$

The first chain consists of functors $(\operatorname{mod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{mod} S^{\mathrm{en}}$; the second consists of functors $(\operatorname{grmod} S^{\mathrm{en}})^{\mathrm{op}} \to \operatorname{grmod} S^{\mathrm{en}}$.

Proof *Ungraded duals.* Non-degeneracy makes $S \to D(S)$, $a \mapsto (s \mapsto \tau(as))$, injective, hence bijective because both spaces have the same finite dimension. Thus for $\lambda \in D(M)$, according to whether M is a right or left module, there exist unique $f(m)$ or $g(m)$ in S satisfying

$$\tau(f(m)s) = \lambda(ms), \qquad \tau(sg(m)) = \lambda(sm) \quad (s \in S,\ m \in M). \tag{1.17}$$

The inverse of $S \to D(S)$ is linear, so $m \mapsto f(m)$ and $m \mapsto g(m)$ are linear. By non-degeneracy, (1.17) gives $f(ms) = f(m)s$ in the right module case and $g(sm) = sg(m)$ in the left module case. Taking

$s = 1$ gives $\alpha_M(f) = \lambda$ or $\beta_M(g) = \lambda$, respectively. Conversely, right or left S-linearity forces any preimage of λ to satisfy the corresponding identity in (1.17). The uniqueness in (1.17) therefore proves that α_M and β_M are bijective. For $s \in S$, $f \in M^*$, and $m \in M$, symmetry and right S-linearity give

$$\alpha_M(sf)(m) = \tau(sf(m)) = \tau(f(m)s) = \tau(f(ms)) = (s\alpha_M(f))(m).$$

Likewise, $\beta_M(gs)(m) = \tau(g(m)s) = \tau(sg(m)) = \tau(g(sm)) = (\beta_M(g)s)(m)$ for a left module M and $g \in {}^*M$. Naturality in (1.14) is routine to check.

Graded duals. For a locally finite graded module M, take the direct sum over $i \in \mathbb{Z}$ of the ungraded comparison maps on M_{-i}. Each summand is an isomorphism, so the resulting maps are isomorphisms of degree 0. Naturality in (1.15) follows by applying the natural isomorphisms (1.14) in each degree.

Two-sided duals. For bimodule compatibility, use (1.11) and

$$\tau(sf(tm)) = \tau(f(tm)s) = \tau(f(tms)), \qquad \tau(g(ms)t) = \tau(tg(ms)) = \tau(g(tms)).$$

The form $\tau_{\mathrm{en}}(a \otimes b^{\mathrm{op}}) = \tau(a)\tau(b)$ is symmetrising on S^{en}. Indeed, $\tau_{\mathrm{en}}((a \otimes b^{\mathrm{op}})(c \otimes d^{\mathrm{op}})) = \tau(ac)\tau(db) = \tau(ca)\tau(bd)$, and its pairing is the tensor product of two non-degenerate pairings. The formula (1.16) satisfies

$$\gamma_M(F)(ms) = \gamma_M(F)(m)s, \qquad \gamma_M(sFt) = s\gamma_M(F)t, \qquad \alpha_M(\gamma_M(F)) = \tau_{\mathrm{en}} \circ F.$$

The argument (1.17), applied to S^{en} and τ_{en}, makes $F \mapsto \tau_{\mathrm{en}} \circ F$ an isomorphism. Thus γ_M is an isomorphism. For a bimodule map $u : M \to N$ and $F \in N^{*_{\mathrm{en}}}$, the formula (1.16) gives $\gamma_M(F \circ u) = \gamma_N(F) \circ u$, proving naturality. Taking the direct sum of $\gamma_{M_{-i}}$ over $i \in \mathbb{Z}$ gives the graded two-sided comparison. Each component is a bimodule isomorphism, and naturality follows in each degree from the ungraded comparison. □

Example 1.51. *The right and left S-duals may consist of different maps to S, even though* (1.13) *identifies both with the linear dual. Let $S = (\Bbbk A_2)_0 = \Bbbk e_1 \oplus \Bbbk e_2$ and $V = (\Bbbk A_2)_1 = \Bbbk e_{12}$, where $e_1 = e_{11}$ and $e_2 = e_{22}$. Use the symmetrising form and linear functional*

$$\begin{aligned} \tau &: S \longrightarrow \Bbbk, \quad ae_1 + be_2 \longmapsto a + b, \\ \varepsilon &: V \longrightarrow \Bbbk, \qquad\quad ce_{12} \longmapsto c \quad (a, b, c \in \Bbbk). \end{aligned}$$

For a right S-linear map $f : V \to S$,

$$f(e_{12}) = f(e_{12}e_2) = f(e_{12})e_2 \in \Bbbk e_2.$$

Requiring $\tau \circ f = \varepsilon$ therefore forces $f(e_{12}) = e_2$. Conversely, the $\Bbbk$-linear map determined by $f(e_{12}) = e_2$ is right S-linear: for $s = ae_1 + be_2 \in S$, we have $f(e_{12}s) = f(be_{12}) = be_2 = f(e_{12})s$.

For a left S-linear map $g : V \to S$,

$$g(e_{12}) = g(e_1e_{12}) = e_1g(e_{12}) \in \Bbbk e_1.$$

Requiring $\tau \circ g = \varepsilon$ instead forces $g(e_{12}) = e_1$. Conversely, the $\Bbbk$-linear map determined by $g(e_{12}) = e_1$ is left S-linear: for $s = ae_1 + be_2 \in S$, we have $g(se_{12}) = g(ae_{12}) = ae_1 = sg(e_{12})$.

The maps are not interchangeable: f is not left S-linear, and g is not right S-linear, since

$$f(e_1e_{12}) = e_2 \neq 0 = e_1f(e_{12}), \qquad g(e_{12}e_2) = e_1 \neq 0 = g(e_{12})e_2.$$

Nevertheless, composing either map with τ gives ε. Thus the comparison maps in (1.13) *satisfy*

$$\alpha_V(f) = \varepsilon = \beta_V(g), \qquad (\beta_V^{-1} \circ \alpha_V)(f) = g \neq f.$$

The isomorphism $V^* \xrightarrow{\sim} {}^*V$ sends f to g; it is not equality of the two spaces of maps inside $\mathrm{Hom}_{\Bbbk}(V,S)$. *Although S is commutative, its left and right actions on V differ.*

Remark 1.52. The ungraded and graded identifications in Theorem 1.50 depend on τ. For instance, $\tau(a_1,\dots,a_r)=\sum_i c_i a_i$ is symmetrising on $\Bbbk^{\oplus r}$ for any $c_i\neq 0$. The theorem compares the different dual constructions separately in the ungraded and graded settings; it does not identify $D(M)$ with $\mathbb{D}(M)$ for an arbitrary infinite-dimensional graded module.

> The two-sided comparison (1.16) also depends on τ; it does not require S^{en} to be semisimple.

For a graded algebra A with $S=A_0$, Theorem 1.50 transports the A-action (1.6) to the graded S-duals of a locally finite graded A-module:

$$af=\alpha_M^{-1}(a\alpha_M(f)) \quad \text{for right } M, \qquad ga=\beta_M^{-1}(\beta_M(g)a) \quad \text{for left } M.$$

Here $\alpha_M(f)=\tau\circ f$ and $\beta_M(g)=\tau\circ g$ are the restrictions of (1.13); $a\in A$, $f\in M^{(*)}$, and $g\in {}^{(*)}M$. The restrictions to S agree with Definition 1.44. These remain duals over S, not duals taking values in A. The graded S-duals are defined without τ, but these transported A-actions depend on τ. The canonical graded A-module is $\mathbb{D}(M)$. Exercise 1.57 computes this dependence.

1.13 Exercises

Exercise 1.53 (Matrix calculations). *Verify the grading on $\Bbbk A_n$ in Example 1.16 using $e_{ij}e_{pq}=\delta_{jp}e_{iq}$. Compute* $\mathrm{rad}(\Bbbk A_3)$ *from the definition.* Hint. *The strictly upper triangular ideal J satisfies $J^3=0$. Apply the argument of Example 1.13 to simple modules and to $A/J\cong\Bbbk^{\oplus 3}$.*

Exercise 1.54 (Shifts and duals). *For $A=\Bbbk A_3$, compute the degrees and all non-zero matrix-unit actions on $\mathbb{D}(e_{22}A)$. Show that it is not isomorphic to any left module Ae_{jj}, even after a shift.* Hint. *Compare dimensions and the actions of e_{11},e_{22},e_{33}.*

Exercise 1.55 (Homogeneous maps). *For the map f in Example 1.26, write every non-zero homogeneous component f_d explicitly and prove that no finite sum equals f.*

Exercise 1.56 (Dependence on the form). *Repeat Example 1.51 with $\tau(e_i)=c_i\neq 0$. For $V=\Bbbk e_{12}$, find the right and left S-linear maps corresponding to the functional $\varepsilon(e_{12})=1$ under* (1.13). Check. *Their values on e_{12} are $c_2^{-1}e_2$ and $c_1^{-1}e_1$, respectively.*

Exercise 1.57 (Dependence of the action on the form). *In this exercise take $\Bbbk=\mathbb{Q}$. Let $A=\Bbbk A_2$, $S=\Bbbk e_{11}\oplus\Bbbk e_{22}$, and $M=e_{11}A$. For $c_1,c_2\in\Bbbk\setminus\{0\}$, define*

$$\tau: S\longrightarrow \Bbbk, \qquad ae_{11}+be_{22}\longmapsto c_1a+c_2b \quad (a,b\in\Bbbk). \tag{1.18}$$

Consider the right S-linear maps

$$\begin{aligned} &\varphi_0: M\longrightarrow S, \quad ae_{11}+be_{12}\longmapsto ae_{11},\\ &\varphi_1: M\longrightarrow S, \quad ae_{11}+be_{12}\longmapsto be_{22} \quad (a,b\in\Bbbk). \end{aligned} \tag{1.19}$$

(1) *Verify that τ is symmetrising and that φ_0,φ_1 form a homogeneous basis of $M^{(*)}$. Determine their degrees.*

(2) *Set $\tau_1(ae_{11}+be_{22})=a+b$ and $\tau_2(ae_{11}+be_{22})=a+2b$. Write $\cdot_1,\cdot_2$ for the respective left A-actions transported from $\mathbb{D}(M)$ using* (1.13). *Calculate the actions of e_{11},e_{22},e_{12} on φ_0,φ_1 for both forms. Are the two actions equal?*

(3) *Construct a graded A-module isomorphism*

$$t:(M^{(*)},\cdot_1)\longrightarrow(M^{(*)},\cdot_2)$$

by giving $t(\varphi_0)$ and $t(\varphi_1)$.

2 Tensor and quadratic algebras

2.1 Tensor products over an algebra

Let A be a graded algebra and set $S = A_0$. Multiplication on A_1 satisfies $(vs)w = v(sw)$ for $v, w \in A_1$ and $s \in S$. A tensor product over S incorporates this identity.

Definition 2.1. *For a $\Bbbk$-algebra S, a right S-module M, and a left S-module N, the tensor product $M \otimes_S N$ is the quotient of $M \otimes_\Bbbk N$ by*

$$\operatorname{span}_\Bbbk\{ms \otimes n - m \otimes sn \mid m \in M,\ n \in N,\ s \in S\}.$$

Write $m \otimes_S n$ for the image of $m \otimes n$. No semisimplicity or finiteness assumption on S, M, N is needed for the following properties.

Lemma 2.2 (Basic properties of tensor products). *Let S be a $\Bbbk$-algebra, $M \in \operatorname{Mod} S$, and $N \in \operatorname{Mod}(S^{\mathrm{op}})$.*

(1) *For a vector space W and a bilinear map $b : M \times N \to W$ satisfying $b(ms, n) = b(m, sn)$ for all $m \in M$, $s \in S$, and $n \in N$, there is a unique linear map $\widetilde{b}$ making the diagram commute:*

$$\begin{array}{ccc} M \times N & \xrightarrow{(m,n)\mapsto m\otimes_S n} & M \otimes_S N \\ & \searrow^{b} & \downarrow{\scriptstyle \exists!\,\widetilde{b}} \\ & & W. \end{array}$$

Conversely, every linear map $M \otimes_S N \to W$ defines such a bilinear map.

(2) *If M, N are S-bimodules, then $M \otimes_S N$ is an S-bimodule with $s(m \otimes_S n)t = sm \otimes_S nt$ for $s, t \in S$.*

(3) *For S-bimodules U, V, W, there are bimodule isomorphisms*

$$\begin{aligned} (U \otimes_S V) \otimes_S W &\xrightarrow{\sim} U \otimes_S (V \otimes_S W), & (u \otimes v) \otimes w &\longmapsto u \otimes (v \otimes w), \\ S \otimes_S V &\xrightarrow{\sim} V, & s \otimes v &\longmapsto sv, \\ V \otimes_S S &\xrightarrow{\sim} V, & v \otimes s &\longmapsto vs. \end{aligned}$$

(4) *Tensor products distribute over direct sums in either variable. For a family $(M_i)_{i\in I}$ of right S-modules, the isomorphism is*

$$\Big(\bigoplus_{i\in I} M_i\Big) \otimes_S N \xrightarrow{\sim} \bigoplus_{i\in I} (M_i \otimes_S N), \qquad (m_i)_i \otimes n \longmapsto (m_i \otimes n)_i.$$

Proof For (1), Definition 1.35 gives a unique linear map $M \otimes_\Bbbk N \to W$ sending $m \otimes n$ to $b(m, n)$. It descends to the quotient precisely when $b(ms, n) - b(m, sn) = 0$. Uniqueness follows because the elements $m \otimes_S n$ span the quotient.

For (2), the outer actions preserve the defining relations: for $a, s, t \in S$,

$$s(ma \otimes n - m \otimes an)t = (sm)a \otimes nt - sm \otimes a(nt).$$

The module axioms follow from those on M, N. In (3), both parenthesisations impose the relations that move an element of S between adjacent factors, so the displayed associativity map and its reverse are well-defined. Their composites fix every $(u \otimes v) \otimes w$ and $u \otimes (v \otimes w)$. The unit inverses are $V \to S \otimes_S V$, $v \mapsto 1 \otimes v$, and $V \to V \otimes_S S$, $v \mapsto v \otimes 1$; the equalities $1 \otimes sv = s \otimes v$ and $vs \otimes 1 = v \otimes s$ show that these maps are inverses.

For (4), the displayed map has finite support because $(m_i)_i$ does. It is well-defined by (1). If $\iota_i : M_i \to \bigoplus_{j\in I} M_j$ is the canonical inclusion, the inverse sends $m \otimes n$ in the ith summand to $\iota_i(m) \otimes n$ and extends linearly over the direct sum. Both composites fix the displayed generators. The same construction works in the other variable. □

Lemma 2.3 (Hom, direct sums, and direct products). *Let M, N, M_i, N_i be right S-modules, with $i \in I$. Write $\iota_i : M_i \to \bigoplus_{j\in I} M_j$ for the inclusions and $\pi_i : \prod_{j\in I} N_j \to N_i$ for the projections. There are natural linear isomorphisms*

$$\operatorname{Hom}_S\Big(\bigoplus_i M_i, N\Big) \xrightarrow{\sim} \prod_i \operatorname{Hom}_S(M_i, N), \quad f \longmapsto (f \circ \iota_i)_i,$$
$$\operatorname{Hom}_S\Big(M, \prod_i N_i\Big) \xrightarrow{\sim} \prod_i \operatorname{Hom}_S(M, N_i), \quad f \longmapsto (\pi_i \circ f)_i.$$

There is also a natural injection

$$\bigoplus_i \operatorname{Hom}_S(M, N_i) \longrightarrow \operatorname{Hom}_S\Big(M, \bigoplus_i N_i\Big), \qquad (f_i)_i \longmapsto \big(m \mapsto (f_i(m))_i\big),$$

which is an isomorphism when M is finitely generated.

Proof The inverse of the first map sends $(f_i)_i$ to $(m_i)_i \mapsto \sum_i f_i(m_i)$; the sum is finite because $(m_i)_i$ has finite support. The inverse of the second sends $(f_i)_i$ to $m \mapsto (f_i(m))_i$; a direct product imposes no support condition. For the third map, coordinate projections recover each f_i, proving injectivity. If $m_1, \dots, m_r$ generate M, the images of these generators under $f : M \to \bigoplus_i N_i$ are supported on a common finite subset $J \subset I$. Then $f(M) \subset \bigoplus_{i\in J} N_i$, so f comes from a finite family of maps. All three constructions commute with precomposition and postcomposition. □

Proposition 2.4 (Tensor products and direct products). *Let S be a $\Bbbk$-algebra, M, M_i right S-modules, and N, N_i left S-modules, with $i \in I$. The coordinate projections define linear maps*

$$\begin{aligned} \kappa_N &: \Big(\prod_i M_i\Big) \otimes_S N \longrightarrow \prod_i (M_i \otimes_S N), \quad (m_i)_i \otimes n \longmapsto (m_i \otimes n)_i, \\ \lambda_M &: M \otimes_S \Big(\prod_i N_i\Big) \longrightarrow \prod_i (M \otimes_S N_i), \quad m \otimes (n_i)_i \longmapsto (m \otimes n_i)_i. \end{aligned} \tag{2.1}$$

(1) *If I is finite, both maps are isomorphisms.*

(2) *The map κ_N is an isomorphism if N is a direct summand of $S^{\oplus r}$ as a left module for some finite r. The corresponding condition on the right module M makes λ_M an isomorphism.*

(3) *If S is finite-dimensional semisimple, both maps are injective. Moreover, κ_N is an isomorphism when N is finite-dimensional, and λ_M is an isomorphism when M is finite-dimensional.*

Proof The formulas in (2.1) are bilinear. For $s \in S$, the component identities $(m_i s) \otimes n = m_i \otimes sn$ and $(ms) \otimes n_i = m \otimes sn_i$ show that both formulas give well-defined maps from the indicated tensor products. For finite I, products equal sums, and Lemma 2.2(4) proves (1).

For $N = S$, the map κ_S is the identity on $\prod_i M_i$ under $m \otimes s \mapsto ms$. For $N = S^{\oplus r}$, both sides are $(\prod_i M_i)^{\oplus r}$, since a finite direct sum commutes with a direct product. If $N \oplus N' \cong S^{\oplus r}$, the comparison for $N \oplus N'$ is $\kappa_N \oplus \kappa_{N'}$. Its bijectivity implies the bijectivity of κ_N. This proves the assertion about κ_N in (2).

Now let S be finite-dimensional semisimple. For finite-dimensional N, Corollary 1.12 gives (2). For arbitrary N, take $z = \sum_{\ell=1}^{r} m^{(\ell)} \otimes n_\ell$ in $\operatorname{Ker} \kappa_N$ and set $N_0 = \sum_{\ell=1}^{r} Sn_\ell$. The submodule

N_0 is finite-dimensional and has a complement in N by Theorem 1.11. Thus the inclusions

$$\Big(\prod_i M_i\Big) \otimes_S N_0 \longrightarrow \Big(\prod_i M_i\Big) \otimes_S N, \qquad M_i \otimes_S N_0 \longrightarrow M_i \otimes_S N$$

are injective, by the direct sum formula in Lemma 2.2(4). Regard z as a tensor with second factors in N_0. The equality $\kappa_N(z) = 0$ then gives $\kappa_{N_0}(z) = 0$. Since κ_{N_0} is an isomorphism, $z = 0$.

For λ_M, apply the same arguments over S^{op}, using $N \otimes_{S^{\mathrm{op}}} M \cong M \otimes_S N$, $n \otimes m \mapsto m \otimes n$. □

The finite-dimensional hypothesis on the fixed factor cannot be dropped from the isomorphism assertion: Example 1.37 gives a non-surjective comparison even for vector spaces. Thus tensor products always distribute over direct sums, but not over arbitrary direct products.

Remark 2.5. For infinite families, Hom need not preserve direct sums in its second variable. For $M = \bigoplus_{i\geq 1} \Bbbk v_i$, the identity $M \to M$ does not come from $\bigoplus_i \mathrm{Hom}_{\Bbbk}(M, \Bbbk v_i)$: its projection onto every $\Bbbk v_i$ is non-zero. Thus the injection in Lemma 2.3 need not be surjective.

A product in the source of Hom behaves differently from a product in the target. For example, a linear map $\prod_{i\geq 1} \Bbbk \to \Bbbk$ need not be determined by its restrictions to the coordinate copies of $\Bbbk$. Indeed, $(1, 1, \ldots) \notin \bigoplus_{i\geq 1} \Bbbk$. Define a functional on $\bigoplus_{i\geq 1} \Bbbk + \Bbbk(1, 1, \ldots)$ to vanish on the direct sum and take value 1 at $(1, 1, \ldots)$, and extend it to the product by extending a basis. The resulting non-zero functional vanishes on every coordinate copy.

For right module maps $u : M \to M'$ and left module maps $v : N \to N'$, the universal property defines

$$u \otimes_S v : M \otimes_S N \longrightarrow M' \otimes_S N', \qquad m \otimes n \longmapsto u(m) \otimes v(n).$$

These maps satisfy

$$(u' \otimes v') \circ (u \otimes v) = (u' \circ u) \otimes (v' \circ v), \qquad \mathrm{id}_M \otimes \mathrm{id}_N = \mathrm{id}_{M \otimes_S N}$$

whenever the maps are composable. Thus, for fixed N and fixed M, respectively, tensor product gives functors

$$(-) \otimes_S N : \mathrm{Mod}\, S \longrightarrow \mathrm{Mod}\, \Bbbk, \qquad M \otimes_S (-) : \mathrm{Mod}(S^{\mathrm{op}}) \longrightarrow \mathrm{Mod}\, \Bbbk.$$

For bimodules and bimodule maps, the same construction preserves the outer actions in Lemma 2.2(2).

The equality $ms \otimes_S n = m \otimes_S sn$ allows a scalar from S to pass between adjacent factors. For an S-bimodule V, define

$$V^{\otimes_S 0} := S, \qquad V^{\otimes_S i} := \underbrace{V \otimes_S \cdots \otimes_S V}_{i \text{ factors}} \quad (i \geq 1).$$

Lemma 2.2 makes these S-bimodules independent of parenthesisation and supplies the identifications used in concatenation.

For an S-bimodule M, the right and left S-duals of Definition 1.44 are themselves S-bimodules, with actions

$$(sft)(m) = sf(tm), \qquad (sgt)(m) = g(ms)t \quad (s, t \in S,\ m \in M). \tag{2.2}$$

Here $f \in M^*$ and $g \in {}^*M$. The duals are ungraded; the parenthesised stars of Definition 1.45 denote different, graded constructions.

Tensor duality reverses the order of the factors. The following maps use only the S-module actions; no basis or symmetrising form is chosen.

Proposition 2.6 (Tensor duality). *Let S be finite-dimensional semisimple and let V, W be finite-dimensional S-bimodules. There are bimodule isomorphisms, natural in V, W,*

$$\begin{aligned} \theta_r &: W^* \otimes_S V^* \xrightarrow{\sim} (V \otimes_S W)^*, \\ \theta_\ell &: {}^*W \otimes_S {}^*V \xrightarrow{\sim} {}^*(V \otimes_S W), \end{aligned} \tag{2.3}$$

defined by

$$\theta_r(g \otimes f)(v \otimes w) = g(f(v)w), \qquad \theta_\ell(g \otimes f)(v \otimes w) = f(vg(w)). \tag{2.4}$$

Convention (Tensor duals). *For S, V, W as in Proposition 2.6, we use θ_r and θ_ℓ in* (2.3) *to identify the dual of a tensor product with the tensor product of the corresponding duals in reverse order. The values of these functionals are always given by* (2.4)*. These are the canonical identifications used below; they do not identify M^*, ${}^*M$, and $D(M)$ with one another. Comparisons with $D(M)$ use a chosen symmetrising form, as in* (1.13)*.*

The proof of Proposition 2.6 is Exercise 2.41.

For the ungraded $\Bbbk$-linear dual D, there is also a natural bimodule isomorphism [GI20, §2.1]

$$\begin{aligned} &\theta_D : D(W) \otimes_S V^* \xrightarrow{\sim} D(V \otimes_S W), \\ &\theta_D(\lambda \otimes f)(v \otimes w) = \lambda(f(v)w). \end{aligned}$$

The actions on $D(W)$ are those in (1.11). For $s \in S$, the equalities $\lambda(f(vs)w) = \lambda(f(v)sw)$ and $(\lambda s)(f(v)w) = \lambda(sf(v)w) = \lambda((sf)(v)w)$ show that the formula gives the same value on $vs \otimes w$ and $v \otimes sw$, and on $\lambda s \otimes f$ and $\lambda \otimes sf$. For $V = S$, the map is the isomorphism $D(W) \otimes_S S \to D(W)$, $\lambda \otimes s \mapsto \lambda s$. The formula commutes with right S-module maps in V, so finite direct sums and direct summands preserve bijectivity; Corollary 1.12 applies to V.

Suppose now that S is equipped with a symmetrising form $\tau : S \to \Bbbk$, as in Definition 1.48. The two-sided dual of Definition 1.47 satisfies

$$\begin{aligned} &\theta_{\mathrm{en}} : W^* \otimes_S V^{*\mathrm{en}} \xrightarrow{\sim} (V \otimes_S W)^{*\mathrm{en}}, \\ &\theta_{\mathrm{en}}(g \otimes F)(v \otimes w) = \sum_j g(a_j w) \otimes b_j^{\mathrm{op}}, \qquad F(v) = \sum_j a_j \otimes b_j^{\mathrm{op}}. \end{aligned} \tag{2.5}$$

We check that the formula defines a map on the two indicated tensor products and preserves both outer actions. For example, $F(vs) = F(v)(s \otimes 1)$ replaces a_j by $a_j s$, while $(Fs)(v) = (1 \otimes s^{\mathrm{op}})F(v)$ replaces b_j by $b_j s$. These give, respectively, compatibility with $vs \otimes w = v \otimes sw$ and with the right outer action. Compatibility with $(gs) \otimes F = g \otimes (sF)$ follows from $(gs)(a_j w) = g(sa_j w)$. The other actions follow from right S-linearity of g and $F(tv) = F(v)(1 \otimes t^{\mathrm{op}})$.

The comparison maps γ_M in (1.16) give

$$\gamma_{V \otimes_S W} \circ \theta_{\mathrm{en}} = \theta_r \circ (\mathrm{id}_{W^*} \otimes \gamma_V),$$

since both sides send $g \otimes F$, evaluated at $v \otimes w$, to $\sum_j g(a_j w)\tau(b_j)$. Thus θ_{en} is an isomorphism by Theorem 1.50. Precomposition shows that θ_D and θ_{en} are natural. No assumption that S^{en} is semisimple is used; compare Remark 1.39.

For the maps $\alpha_M(f) = \tau \circ f$ in (1.13),

$$\alpha_{V \otimes_S W} \circ \theta_r = \theta_D \circ (\alpha_W \otimes \mathrm{id}_{V^*}),$$

because both sides send $g \otimes f$ to the functional $v \otimes w \mapsto \tau(g(f(v)w))$.

Example 2.7. *Let $S = \Bbbk^{\oplus r}$ with coordinate idempotents $e_1, \ldots, e_r$, and suppose $V = e_i V e_j$ and*

$W = e_j W e_\ell$. *By* (2.2),

$$V \otimes_S W = e_i(V \otimes_S W)e_\ell, \qquad W^* \otimes_S V^* = e_\ell(W^* \otimes_S V^*)e_i.$$

Thus θ_r *in* (2.3) *reverses the two outer idempotents.*

2.2 Generated algebras

Let A be graded with $S = A_0$. Multiplication makes A_1 an S-bimodule and gives

$$\mu_i : A_1^{\otimes_S i} \longrightarrow A_i, \qquad a_1 \otimes \cdots \otimes a_i \longmapsto a_1 \cdots a_i \quad (i \geq 1), \qquad \mu_0 = \mathrm{id}_S\,. \tag{2.6}$$

Definition 2.8. *The algebra* A *is generated in degree* 1 *over* S *if every* μ_i *in* (2.6) *is surjective.*

Equivalently, S and A_1 generate A as an algebra.

Example 2.9. *The algebra* $\Bbbk A_n$ *of Example 1.16 is generated in degree* 1 *over* $S = \bigoplus_i \Bbbk e_{ii}$*:*

$$e_{ij} = e_{i,i+1}e_{i+1,i+2}\cdots e_{j-1,j} \quad (1 \leq i < j \leq n).$$

For the grading $\deg e_{ij} = 2(j-i)$*, the same algebra has* $A_1 = 0$ *and* $A_2 \neq 0$*, so it is not generated in degree* 1*.*

Example 2.10. *The polynomial algebra* $\Bbbk[x_1, \ldots, x_n]$*, with each* x_i *in degree* 1*, is generated in degree* 1 *over* $\Bbbk$*. Indeed, every monomial of total degree* d *is a product of* d *variables, and these monomials span the degree* d *component.*

Example 2.11. *Let* A *be the algebra of upper triangular* 2×2 *matrices over* $\Bbbk[x]$*, graded by* $\deg(x^m e_{ij}) = m + j - i$ *for* $m \geq 0$ *and* $1 \leq i \leq j \leq 2$*. Then*

$$A_0 = \Bbbk e_{11} \oplus \Bbbk e_{22}, \qquad A_d = \Bbbk x^d e_{11} \oplus \Bbbk x^d e_{22} \oplus \Bbbk x^{d-1} e_{12} \quad (d \geq 1).$$

It is generated in degree 1 *over* A_0*, since*

$$x^d e_{11} = (xe_{11})^d, \qquad x^d e_{22} = (xe_{22})^d, \qquad x^{d-1}e_{12} = (xe_{11})^{d-1}e_{12} \quad (d \geq 1).$$

Here A_0 *has two simple summands,* $A_d \neq 0$ *for every* $d \geq 0$*, and* A *is non-commutative:* $e_{11}e_{12} = e_{12}$ *whereas* $e_{12}e_{11} = 0$*.*

2.3 Tensor algebras and the universal property

Definition 2.12. *Let* S *be a* $\Bbbk$*-algebra and* V *an* S*-bimodule. The tensor algebra of* V *over* S *is*

$$T_S(V) := \bigoplus_{i \in \mathbb{Z}} T_S(V)_i, \quad \text{where} \quad T_S(V)_i = \begin{cases} V^{\otimes_S i}, & i \geq 1, \\ S, & i = 0, \\ 0, & i < 0, \end{cases}$$

with multiplication given by concatenation of tensors and the given S*-actions.*

Every element of $T_S(V)$ is a finite sum of tensors, since the grading is a direct sum. The unit is $1_S \in T_S(V)_0$. Associativity follows from the associativity and unit isomorphisms in Lemma 2.2(3). The left and right S-actions determine which factors can be concatenated. Juxtaposition $v_1 \cdots v_d$ in $T_S(V)$ denotes $v_1 \otimes_S \cdots \otimes_S v_d$; in a specified quotient it denotes the image of that tensor.

Example 2.13 (Polynomial and exterior algebras). *Let V, W be finite-dimensional vector spaces with bases $x_1, \dots, x_n$ and $y_1, \dots, y_m$, respectively. The symmetric algebra of V and the exterior algebra of W are*

$$\begin{aligned} \operatorname{Sym}(V) &= T_{\Bbbk}(V)/(x_i x_j - x_j x_i \mid i < j) \cong \Bbbk[x_1, \dots, x_n], \\ \bigwedge W &= T_{\Bbbk}(W)/(w^2 \mid w \in W). \end{aligned} \tag{2.7}$$

Both algebras have degree 0 component $\Bbbk$ and are generated in degree 1, with degree 1 components V and W, respectively. The polynomial basis allows repetitions; the exterior basis consists of 1 and $y_{j_1} \cdots y_{j_d}$ with $j_1 < \cdots < j_d$. The exterior relations are equivalently $y_j^2 = 0$ for $1 \le j \le m$ and $y_i y_j + y_j y_i = 0$ for $i < j$, in every characteristic.

The bases in Example 2.13 can be checked directly. For $\operatorname{Sym}(V)$, commuting adjacent generators gives the ordered monomials, which are independent under the map to the polynomial ring. For $\bigwedge W$, the relations reduce every word to a multiple of a strictly increasing word or to zero. To prove independence in degree d, fix $j_1 < \cdots < j_d$ and use the linear functional $W^{\otimes_{\Bbbk} d} \to \Bbbk$ sending $w_1 \otimes \cdots \otimes w_d$ to the determinant of their coordinates in rows $j_1, \dots, j_d$ relative to the basis $y_1, \dots, y_m$. The determinant vanishes on tensors with two equal adjacent factors, hence on the degree d part of $(w^2 \mid w \in W)$. On increasing basis words it is 1 on $y_{j_1} \cdots y_{j_d}$ and 0 on the others, proving independence in every characteristic.

In Lemma 2.14, only components with matching local units contribute to the tensor product.

Lemma 2.14. *Let $S = \bigoplus_{i=1}^{r} \Bbbk e_i$ with coordinate idempotents and let V, W be S-bimodules. There is a bimodule isomorphism*

$$V \otimes_S W \cong \bigoplus_{i,j,\ell} e_i V e_j \otimes_{\Bbbk} e_j W e_\ell. \tag{2.8}$$

Consequently, after choosing a basis of each $e_i V e_j$, a basis of $V^{\otimes_S d}$ for $d \ge 1$ consists of the words of length d whose adjacent idempotents match.

Proof The mutually orthogonal idempotents sum to 1, so

$$V = \bigoplus_{i,j} e_i V e_j, \qquad (e_i V e_j) \otimes_S (e_p W e_\ell) = 0 \quad \text{if } j \ne p. \tag{2.9}$$

For the second identity, $v \otimes w = v e_j \otimes w = v \otimes e_j w = 0$. The inclusions of the matching components define the map from right to left in (2.8). Its inverse is

$$V \otimes_S W \longrightarrow \bigoplus_{i,j,\ell} e_i V e_j \otimes_{\Bbbk} e_j W e_\ell, \qquad v \otimes_S w \longmapsto \sum_{i,j,\ell} e_i v e_j \otimes_{\Bbbk} e_j w e_\ell.$$

For $s = \sum_j s_j e_j \in S$, replacing (v, w) by (vs, w) or (v, sw) multiplies the summands with middle index j by the same scalar s_j, so the inverse is well-defined. One composite sends $v \otimes_S w$ to $\sum_{i,j,\ell} e_i v e_j \otimes_S e_j w e_\ell = v \otimes_S w$. On $v \otimes_{\Bbbk} w$ with $v \in e_i V e_j$ and $w \in e_j W e_\ell$, the other composite has only the (i, j, ℓ) component, equal to $v \otimes_{\Bbbk} w$. Iterating (2.8) and using the basis construction following Definition 1.35 proves the assertion about words. □

Example 2.15. *We apply Lemma 2.14 to realise $\Bbbk A_3$ as a tensor algebra over its diagonal subalgebra. Let $S = \Bbbk^{\oplus 3}$ with coordinate idempotents e_1, e_2, e_3, and let $V = \Bbbk a \oplus \Bbbk b$ have actions $e_1 a e_2 = a$ and $e_2 b e_3 = b$, with all other idempotent actions zero. By Lemma 2.14,*

$$V \otimes_S V = \Bbbk(a \otimes b), \qquad V^{\otimes_S 3} = 0, \qquad T_S(V) = S \oplus \Bbbk a \oplus \Bbbk b \oplus \Bbbk ab.$$

Here $ab = a \otimes b$. The assignments $e_i \mapsto e_{ii}$, $a \mapsto e_{12}$, and $b \mapsto e_{23}$ give $T_S(V) \cong \Bbbk A_3$; the tensor ab maps to e_{13}.

Proposition 2.16 (Universal property of the tensor algebra). *Let $\iota_S : S \to T_S(V)$ and $\iota_V : V \to T_S(V)$ be the canonical inclusions. Given a unital $\Bbbk$-algebra homomorphism $\varphi : S \to B$ and an S-bimodule map $f : V \to B$, where $sbt = \varphi(s)b\varphi(t)$ on B, there is a unique $\Bbbk$-algebra homomorphism $\widetilde{f} : T_S(V) \to B$ making the diagram commute:*

$$\begin{array}{ccccc} V & \xrightarrow{\iota_V} & T_S(V) & \xleftarrow{\iota_S} & S \\ & \searrow^{f} & \downarrow \exists!\, \widetilde{f} & \swarrow_{\varphi} & \\ & & B. & & \end{array} \tag{2.10}$$

Proof Define $\widetilde{f}$ on S and each tensor power by

$$\widetilde{f}(s) = \varphi(s), \qquad \widetilde{f}(v_1 \otimes_S \cdots \otimes_S v_i) = f(v_1)\cdots f(v_i) \quad (s \in S,\ i \geq 1). \tag{2.11}$$

The identity $f(vs)f(w) = f(v)\varphi(s)f(w) = f(v)f(sw)$ shows that replacing adjacent factors $vs \otimes w$ by $v \otimes sw$ does not change the value in (2.11). Since every element of $T_S(V)$ has finitely many non-zero components, these formulas extend to a linear map $T_S(V) \to B$. Concatenation of tensors becomes multiplication in B, and the identities $f(sv) = \varphi(s)f(v)$ and $f(vs) = f(v)\varphi(s)$ handle the degree 0 factors. Also, $\widetilde{f}(1_S) = \varphi(1_S) = 1_B$. Every algebra homomorphism making (2.10) commute must have the values in (2.11), proving uniqueness. □

If B is graded, $\varphi(S) \subset B_0$, and $f(V) \subset B_1$, then (2.11) gives $\widetilde{f}(V^{\otimes_S i}) \subset B_i$. Thus $\widetilde{f}$ is a graded algebra homomorphism.

Corollary 2.17. *If A is generated in degree 1 over $S = A_0$, there is a surjective graded algebra homomorphism $\mu : T_S(A_1) \twoheadrightarrow A$ whose degree i component is μ_i from (2.6).*

Proof Apply Proposition 2.16 to the inclusions $S \to A$ and $A_1 \to A$. Surjectivity of each μ_i gives surjectivity of μ. □

Definition 2.18. *For the map μ in Corollary 2.17, the homogeneous two-sided ideal $I := \operatorname{Ker} \mu$ is the ideal of relations.*

Writing $V = A_1$, the components $\mu_0 = \mathrm{id}_S$ and $\mu_1 = \mathrm{id}_V$ give $I_0 = I_1 = 0$. Since μ preserves degrees, its kernel is homogeneous, and

$$T_S(V)/I \xrightarrow{\sim} A, \qquad z + I \longmapsto \mu(z)$$

is a graded algebra isomorphism. The bimodule V specifies the generators and their left and right S-actions; I specifies the relations between their products. Homogeneity is essential for the quotient grading: the relation $x - x^2$ in Example 1.33 does not satisfy this condition.

2.4 Quadratic algebras

We now consider algebras whose ideal of relations is generated in degree 2.

Definition 2.19. *A graded algebra A is quadratic if there exist a finite-dimensional semisimple $\Bbbk$-algebra S, a finite-dimensional S-bimodule V, and an S-sub-bimodule $R \subset V \otimes_S V$ such that $A \cong T_S(V)/(R)$ as graded algebras, where V is in degree 1 and (R) is the two-sided ideal of $T_S(V)$ generated by R.*

An isomorphism $A \cong T_S(V)/(R)$ as in Definition 2.19 is called a quadratic presentation of A.

Proposition 2.20. *A quadratic algebra $A = T_S(V)/(R)$ has $A_0 = S$, $A_1 = V$, and $A_2 = (V \otimes_S V)/R$, is locally finite, and is generated in degree 1 over S.*

Proof The ideal (R) has no components in degrees 0 and 1. Its degree 2 component is exactly R. Each A_i is a quotient of the finite-dimensional space $V^{\otimes_S i}$, and multiplication maps $V^{\otimes_S i}$ onto A_i. □

Thus a quadratic presentation recovers $S \cong A_0$, $V \cong A_1$, and $R \cong \operatorname{Ker}(A_1 \otimes_{A_0} A_1 \to A_2)$. In particular, the kernel of the canonical map $T_{A_0}(A_1) \twoheadrightarrow A$ is generated by its degree 2 component. Quadraticity depends on the grading: for $n \geq 2$, the grading $\deg e_{ij} = 2(j-i)$ on $\Bbbk A_n$ is not quadratic, since it has no degree 1 generators.

Example 2.21. *Let $S = \bigoplus_{i=1}^{n} \Bbbk e_{ii}$ and $V = \bigoplus_{i=1}^{n-1} \Bbbk e_{i,i+1}$ inside $\Bbbk A_n$. By Lemma 2.14, the basis tensors of positive length are*

$$e_{i,i+1} \otimes_S e_{i+1,i+2} \otimes_S \cdots \otimes_S e_{j-1,j} \longmapsto e_{ij} \quad (1 \leq i < j \leq n).$$

Thus multiplication gives $T_S(V) \cong \Bbbk A_n$, a quadratic algebra with $R = 0$.

Example 2.22. *For S, V, a, b from Example 2.15, the quotient*

$$T_S(V)/(ab) \cong \Bbbk A_3/(e_{13})$$

is quadratic with $R = \Bbbk(a \otimes b)$. Its basis is e_1, e_2, e_3, a, b, in degrees $0, 0, 0, 1, 1$, as in Example 1.32.

Example 2.23. *Let $S = \Bbbk^{\oplus 2}$ with coordinate idempotents e_1, e_2, and let $V = \Bbbk a \oplus \Bbbk b$ satisfy $e_1 a e_2 = a$ and $e_2 b e_1 = b$, with all other idempotent actions zero. By Lemma 2.14, the alternating words in a, b form the basis of positive degree tensors in $T_S(V)$. The quadratic algebra $A = T_S(V)/(ab)$ has*

$$A = S \oplus \Bbbk a \oplus \Bbbk b \oplus \Bbbk ba, \qquad \dim_{\Bbbk} A_0 = 2, \quad \dim_{\Bbbk} A_1 = 2, \quad \dim_{\Bbbk} A_2 = 1.$$

Every alternating word of length at least 3 contains ab, so all higher components vanish.

Example 2.24. *Let $B = \Bbbk A_4$ and $A = B/(e_{14})$, with the grading of Example 1.16. The ideal $I = (e_{14}) = \Bbbk e_{14}$ has $I_2 = 0$ but $I_3 \neq 0$. Since $e_{14} = e_{12} e_{23} e_{34}$, the algebra A is generated in degree 1 with a relation of degree 3 and no relations of degree 2. By Proposition 2.20, any quadratic presentation would have the same generators and zero quadratic relation space, hence would give $\Bbbk A_4$ itself. Thus A is not quadratic.*

Example 2.25. *The symmetric and exterior algebras of Example 2.13 are quadratic: the defining relations in (2.7) belong to $V \otimes_{\Bbbk} V$ and $W \otimes_{\Bbbk} W$, respectively.*

2.5 Hilbert series

Fix a finite-dimensional semisimple $\Bbbk$-algebra S, concentrated in degree 0. We record dimensions relative to all simple right S-modules. For an S-bimodule, we retain both the left and the right indices.

Definition 2.26. *Choose representatives $L_1, \ldots, L_r$ of the isomorphism classes of simple right S-modules. For a finite-dimensional right S-module M, write, with $m_j \in \mathbb{Z}_{\geq 0}$,*

$$M \cong \bigoplus_{j=1}^{r} L_j^{\oplus m_j}, \qquad \mathbf{dim} M := (m_1, \ldots, m_r).$$

The row vector $\mathbf{dim} M$ is the dimension vector of M. For a finite-dimensional S-bimodule U, we record its dimension vector with two indices, whose rows are the dimension vectors of $L_i \otimes_S U$ for $1 \leq i \leq r$:

$$\mathbf{dim} U = (u_{ij})_{1 \leq i,j \leq r}, \qquad L_i \otimes_S U \cong \bigoplus_{j=1}^{r} L_j^{\oplus u_{ij}}.$$

Here $u_{ij} \in \mathbb{Z}_{\geq 0}$, and row i is $\mathbf{dim}(L_i \otimes_S U)$.

The decompositions exist by Theorem 1.11. Their multiplicities are unique: a non-zero homomorphism between simple modules is an isomorphism, since its kernel and image are submodules. Consequently

$$m_j = \frac{\dim_{\Bbbk} \operatorname{Hom}_S(L_j, M)}{\dim_{\Bbbk} \operatorname{End}_S(L_j)},$$

and the same formula applies to each $L_i \otimes_S U$. In particular, the dimension data do not depend on choices of decompositions. For the S-bimodule S, we have $\mathbf{dim} S = I_r$, the $r \times r$ identity array.

Example 2.27. *If $S = \bigoplus_{i=1}^{r} \Bbbk e_i$ with coordinate idempotents, take $L_i = e_i S$. Then*

$$\mathbf{dim} M = (\dim_{\Bbbk} M e_j)_{j=1}^{r}, \qquad (\mathbf{dim} U)_{ij} = \dim_{\Bbbk}(e_i U e_j).$$

Indeed, $L_i \otimes_S U \cong e_i U$ by multiplication, and the multiplicity of $e_j S$ in $e_i U$ is $\dim_{\Bbbk}(e_i U e_j)$. For $S = \Bbbk$, the dimension vector has one entry, the usual vector-space dimension.

Definition 2.28. *Let $M = \bigoplus_{d \in \mathbb{Z}} M_d$ be a locally finite graded right S-module, with $M_d = 0$ for all sufficiently negative d. Its Hilbert series is*

$$H_M(t) := \sum_{d \in \mathbb{Z}} (\mathbf{dim} M_d) t^d.$$

For a locally finite graded S-bimodule U, bounded below in the same sense, define $H_U(t)$ by the same formula using the two-index dimension data of Definition 2.26. For a locally finite graded algebra A with $A_0 = S$, its Hilbert series $H_A(t)$ uses the S-bimodule structure on each A_d. In particular, $H_A(0) = I_r$ and, when $S = \bigoplus_i \Bbbk e_i$,

$$H_A(t)_{ij} = \sum_{d \geq 0} \dim_{\Bbbk}(e_i A_d e_j) t^d.$$

The entries are formal Laurent series with integer coefficients; for algebras they are formal power series. A finite-dimensional graded module or algebra gives a Laurent polynomial or polynomial, respectively. No numerical convergence is required: a coefficient is specified degree by degree. This is one definition of Hilbert series, with the appropriate dimension data for a right module or a bimodule. For an algebra, keeping both indices records where products can be formed.

Lemma 2.29. *For locally finite graded right S-modules, or graded S-bimodules, bounded below:*

(1) *A graded isomorphism $M \cong N$ implies $H_M(t) = H_N(t)$.*

(2) *A short exact sequence $0 \to L \to M \to N \to 0$ gives $H_M(t) = H_L(t) + H_N(t)$.*

(3) *For every $a \in \mathbb{Z}$, $H_{M(a)}(t) = t^{-a} H_M(t)$.*

Proof A graded isomorphism preserves each degree and its simple-module multiplicities. A short exact sequence of right S-modules splits by semisimplicity, so multiplicities add. For bimodules, apply $L_i \otimes_S (-)$ in each degree: the original sequence splits as a sequence of left S-modules, so its tensor sequence is exact, and right-module multiplicities again add. Finally, $M(a)_d = M_{d+a}$ gives

$$H_{M(a)}(t) = \sum_d (\mathbf{dim} M_{d+a}) t^d = t^{-a} \sum_j (\mathbf{dim} M_j) t^j. \qquad \square$$

Definition 2.30. *For graded right and left S-modules M, N, their tensor product has total grading*

$$(M \otimes_S N)_d = \bigoplus_{p+q=d} M_p \otimes_S N_q.$$

The same definition applies to bimodules, and to vector spaces by taking $S = \Bbbk$.

The decomposition follows by distributing tensor products over direct sums, as in Lemma 2.2(4). When M, N are locally finite and bounded below, only finitely many pairs (p, q) contribute in a fixed degree, so the tensor product is locally finite and bounded below.

To multiply two arrays $C = (c_{ij})$ and $D = (d_{ij})$, or a row vector $m = (m_i)$ and an array C, use

$$(CD)_{i\ell} = \sum_{j=1}^{r} c_{ij} d_{j\ell}, \qquad (mC)_j = \sum_{i=1}^{r} m_i c_{ij}.$$

Thus multiplication sums over the matching middle index, rather than multiplying individual entries independently. For series, use the same products on coefficients and collect terms with the same total power of t.

Lemma 2.31. *Let U, W be finite-dimensional S-bimodules and M a finite-dimensional right S-module. Then*

$$\mathbf{dim}(U \otimes_S W) = (\mathbf{dim} U)(\mathbf{dim} W), \qquad \mathbf{dim}(M \otimes_S U) = (\mathbf{dim} M)(\mathbf{dim} U).$$

For locally finite graded modules with the indicated actions, bounded below, this gives

$$H_{U \otimes_S W}(t) = H_U(t) H_W(t), \qquad H_{M \otimes_S U}(t) = H_M(t) H_U(t).$$

Proof Write $C = \mathbf{dim} U$ and $D = \mathbf{dim} W$. Associativity and distribution over direct sums give

$$\begin{aligned} L_i \otimes_S (U \otimes_S W) &\cong (L_i \otimes_S U) \otimes_S W \\ &\cong \bigoplus_j (L_j \otimes_S W)^{\oplus c_{ij}} \cong \bigoplus_\ell L_\ell^{\oplus \sum_j c_{ij} d_{j\ell}}. \end{aligned}$$

This proves the first equality. Decomposing $M \cong \bigoplus_i L_i^{\oplus m_i}$ proves the second in the same way. Apply these equalities to each $U_p \otimes_S W_q$ or $M_p \otimes_S U_q$ and sum over $p + q = d$ to obtain the series identities. □

Example 2.32 (Hilbert series of a tensor algebra). *Let V be a finite-dimensional S-bimodule, placed in degree 1, and put $C = \mathbf{dim} V$. Since $T_S(V)_d = V^{\otimes_S d}$ and $V^{\otimes_S 0} = S$, Lemma 2.31 gives*

$$\mathbf{dim} T_S(V)_d = C^d \quad (d \geq 0), \qquad C^0 = I_r.$$

Consequently,

$$\boxed{H_{T_S(V)}(t) = \sum_{d \geq 0} t^d C^d = (I_r - tC)^{-1} = (I_r - t\,\mathbf{dim} V)^{-1}.} \tag{2.12}$$

Indeed, multiplying the series on either side by $I_r - tC$ gives constant coefficient I_r; the coefficient of t^d for $d \geq 1$ is $C^d - C^d = 0$. The inverse is therefore an identity of formal power series with the coefficient multiplication defined above.

For $S = \Bbbk$ and $\dim_\Bbbk V = m$, the formula is $H_{T_\Bbbk(V)}(t) = 1/(1 - mt)$. More generally, for $S = \mathrm{Mat}_n(\Bbbk)$ and $V = S^{\oplus m}$ as an S-bimodule, there is one simple right S-module $L = \Bbbk^{1 \times n}$. Since $L \otimes_S V \cong L^{\oplus m}$, we again have $C = (m)$ and $H_{T_S(V)}(t) = 1/(1 - mt)$. Here the coefficient records simple-module multiplicities, not $\dim_\Bbbk V = mn^2$.

Example 2.33 (Upper triangular algebras and a quotient). *For $A = \Bbbk A_n$, order the coordinate idempotents as $e_{11}, \ldots, e_{nn}$. Let C have $C_{i,i+1} = 1$ for $1 \leq i < n$ and all other entries zero. The tensor presentation in Example 2.21 and (2.12) give*

$$H_A(t) = I_n + tC + \cdots + t^{n-1} C^{n-1}, \qquad H_A(t)_{ij} = \begin{cases} t^{j-i}, & i \leq j, \\ 0, & i > j. \end{cases}$$

For $n = 3$, killing e_{13} removes the degree 2 contribution in entry $(1,3)$:

$$H_{\Bbbk A_3}(t) = \begin{pmatrix} 1 & t & t^2 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}, \qquad H_{\Bbbk A_3/(e_{13})}(t) = \begin{pmatrix} 1 & t & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}.$$

This also follows from additivity in Lemma 2.29.

Example 2.34 (Two alternating generators). *For S, V, a, b in Example 2.23, use the order e_1, e_2. Then*

$$C = \mathbf{dim} V = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad C^2 = I_2, \qquad H_{T_S(V)}(t) = \frac{1}{1-t^2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

In $A = T_S(V)/(ab)$, the degree 2 basis vector ba belongs to $e_2 A e_2$, so

$$H_A(t) = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}.$$

Example 2.35. *For the algebras in Example 2.13, $S = \Bbbk$ and there is one dimension entry. Each polynomial generator may occur any number of times, whereas each exterior generator occurs at most once. Thus*

$$H_{\mathrm{Sym}(V)}(t) = \frac{1}{(1-t)^n}, \qquad H_{\bigwedge W}(t) = (1+t)^m \quad \textit{where } n = \dim_{\Bbbk} V,\ m = \dim_{\Bbbk} W.$$

Lemma 2.36. *For locally finite graded vector spaces U, W bounded below, $H_{U\otimes_{\Bbbk} W}(t) = H_U(t) H_W(t)$; this is the case $S = \Bbbk$ of Lemma 2.31. More generally, let U be a graded $\Bbbk^{\oplus r}$-bimodule and W a graded $\Bbbk^{\oplus s}$-bimodule, locally finite and bounded below. Write $e_1, \dots, e_r$ and $f_1, \dots, f_s$ for their coordinate idempotents. As a bimodule over $\Bbbk^{\oplus r} \otimes_{\Bbbk} \Bbbk^{\oplus s} \cong \Bbbk^{\oplus rs}$, with idempotents $e_i \otimes f_p$, the total tensor product satisfies*

$$H_{U\otimes_{\Bbbk} W}(t)_{(i,p),(j,q)} = H_U(t)_{ij} H_W(t)_{pq} \quad (1 \le i, j \le r,\ 1 \le p, q \le s).$$

Proof The second formula follows from

$$(e_i \otimes f_p)(U_a \otimes_{\Bbbk} W_b)(e_j \otimes f_q) = (e_i U_a e_j) \otimes_{\Bbbk} (f_p W_b f_q)$$

by multiplying vector-space dimensions and summing over $a + b = d$ in degree d. □

2.6 The degree 2 relations of a quotient

Let S be finite-dimensional semisimple, V a finite-dimensional S-bimodule, and $I \subset T_S(V)$ a homogeneous two-sided ideal with $I_0 = I_1 = 0$. Set $A = T_S(V)/I$. In degree 2,

$$A_2 \cong (V \otimes_S V)/I_2, \quad \text{where} \quad I_2 = I \cap (V \otimes_S V).$$

Definition 2.37. *The quadratic part of A is $qA := T_S(V)/(I_2)$.*

Since $(I_2) \subset I$, there is a surjective graded algebra homomorphism

$$\pi_A : qA \longrightarrow A, \qquad z + (I_2) \longmapsto z + I, \qquad \operatorname{Ker} \pi_A = I/(I_2).$$

Consequently,

$$\pi_A \text{ is an isomorphism} \iff I = (I_2) \iff A \text{ is quadratic.}$$

For the last equivalence, (I_2) gives a quadratic presentation, while Proposition 2.20 identifies the generators and degree 2 relations of any quadratic presentation with A_1 and I_2.

Example 2.38. *For $A = \Bbbk A_4/(e_{14})$ from Example 2.24, we have $I_2 = 0$, so*

$$qA = \Bbbk A_4, \qquad \pi_A : \Bbbk A_4 \twoheadrightarrow \Bbbk A_4/(e_{14}), \qquad \operatorname{Ker} \pi_A = \Bbbk e_{14}.$$

Thus qA does not impose the cubic relation $e_{12}e_{23}e_{34} = 0$.

2.7 Orthogonal complements and quadratic duality

Let S be finite-dimensional semisimple.

Definition 2.39. *For an S-sub-bimodule U of a finite-dimensional S-bimodule M, its right orthogonal complement and left orthogonal complement are*

$$U^{\perp} := \{f \in M^* \mid f(u) = 0 \text{ for all } u \in U\},$$
$${}^{\perp}U := \{g \in {}^*M \mid g(u) = 0 \text{ for all } u \in U\}.$$

Both orthogonal complements are S-sub-bimodules. Indeed, for $s, t \in S$ and $u \in U$, (2.2) gives

$$(sft)(u) = sf(tu) = 0 \quad (f \in U^{\perp}), \qquad (sgt)(u) = g(us)t = 0 \quad (g \in {}^{\perp}U),$$

since $tu, us \in U$. The orthogonal complements are defined without choosing a symmetrising form.

If S is equipped with a symmetrising form $\tau : S \to \Bbbk$, the comparison maps α_M, β_M of (1.13) satisfy

$$\alpha_M(U^{\perp}) = \beta_M({}^{\perp}U) = \{\lambda \in D(M) \mid \lambda(u) = 0 \text{ for all } u \in U\}. \tag{2.13}$$

The forward inclusions follow by composing with τ. Conversely, $\alpha_M(f)(U) = 0$ gives

$$\tau(f(u)s) = \tau(f(us)) = 0 \quad \text{for all } s \in S,\ u \in U.$$

Non-degeneracy of τ forces $f(u) = 0$ for every $u \in U$. For the left dual, $\tau(sg(u)) = \tau(g(su)) = 0$ gives $g(u) = 0$ instead. Bijectivity of α_M, β_M in Theorem 1.50 proves (2.13).

Example 2.40. *We compute the orthogonal complements of the symmetric and exterior relation spaces separately. Take $V = \Bbbk x_1 \oplus \Bbbk x_2$ and $W = \Bbbk y_1 \oplus \Bbbk y_2$ in Example 2.13. The relation spaces in $V \otimes_{\Bbbk} V$ and $W \otimes_{\Bbbk} W$, respectively, are*

$$R_{\mathrm{Sym}(V)} = \Bbbk(x_1 \otimes x_2 - x_2 \otimes x_1),$$
$$R_{\bigwedge W} = \Bbbk(y_1 \otimes y_1) \oplus \Bbbk(y_2 \otimes y_2) \oplus \Bbbk(y_1 \otimes y_2 + y_2 \otimes y_1).$$

Let x_1^*, x_2^* and y_1^*, y_2^* be the dual bases of $D(V)$ and $D(W)$, respectively. Under the tensor-dual identification (2.4),

$$R_{\mathrm{Sym}(V)}^{\perp} = \operatorname{span}_{\Bbbk}\{x_1^* \otimes x_1^*,\ x_2^* \otimes x_2^*,\ x_1^* \otimes x_2^* + x_2^* \otimes x_1^*\},$$
$$R_{\bigwedge W}^{\perp} = \Bbbk(y_1^* \otimes y_2^* - y_2^* \otimes y_1^*).$$

The first space consists of the exterior relations on $D(V)$; the second consists of the symmetric relations on $D(W)$.

2.8 Exercises

Exercise 2.41 (Interaction of tensor product with S-dual). *For S, V, W as in Proposition 2.6, prove the assertion about right S-duals.*

(1) *Check that the formula for θ_r in (2.4) is bilinear in each pair of variables, is unchanged by*

$$vs \otimes w \longleftrightarrow v \otimes sw, \qquad gs \otimes f \longleftrightarrow g \otimes sf \quad (s \in S),$$

and gives a right S-linear map $V \otimes_S W \to S$ for each $g \otimes f$. Verify $\theta_r(sg \otimes ft) = s\theta_r(g \otimes f)t$ for $s, t \in S$.

(2) *Prove that θ_r is bijective.*

(3) *For bimodule maps $p : V \to V'$ and $q : W \to W'$, prove*

$$(p \otimes q)^* \circ \theta_r^{V',W'} = \theta_r^{V,W} \circ (q^* \otimes p^*).$$

Here the superscripts specify the pair of bimodules, and p^*, q^*, and $(p \otimes q)^*$ are given by precomposition with p, q, and $p \otimes q$, respectively.

Hint for (2). *First take $V = S$, then use Corollary 1.12.*

Remark. *The analogous assertions hold for left S-duals, with left and right interchanged.*

Exercise 2.42. *Let $A = \Bbbk A_4/(e_{13}, e_{24})$ with the grading of Example 1.16. Find a homogeneous basis, compute $H_A(t)$, and write $A = T_S(V)/(R)$ with explicit S, V, R.* Hint. *Use Lemma 2.14; the ideal contains e_{14} as well as e_{13}, e_{24}.*

Exercise 2.43. *For the bimodule V in Example 2.23, find a basis of each $T_S(V)_d$ and show that*

$$H_{T_S(V)}(t) = \frac{1}{1-t^2}\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}.$$

Compare with the quotient $T_S(V)/(ab)$. Hint. *Count alternating words separately in each degree, including degree 0.*

Exercise 2.44. *For the algebras in Example 2.13, prove*

$$\dim_{\Bbbk} \mathrm{Sym}(V)_2 = \frac{n(n+1)}{2}, \qquad \dim_{\Bbbk}(\bigwedge W)_2 = \frac{m(m-1)}{2} \quad \text{where } n = \dim_{\Bbbk} V,\ m = \dim_{\Bbbk} W.$$

Hint. *Count the ordered pairs allowed in the two bases of Example 2.13.*

3 The quadratic dual

3.1 Generators and orthogonal relations

Fix a quadratic presentation $A = T_S(V)/(R)$ as in Definition 2.19: S is finite-dimensional semisimple, V is a finite-dimensional S-bimodule, and $R \subset V \otimes_S V$ is an S-sub-bimodule. We will replace the generators by an S-dual and the relations by the S-linear maps that vanish on R. Use $V^* = \mathrm{Hom}_S(V,S)$ and ${}^*V = \mathrm{Hom}_{S^{\mathrm{op}}}(V,S)$ from Definition 1.44, with the bimodule actions in (2.2). The tensor isomorphisms (2.3) have values

$$\begin{aligned} \theta_r(g \otimes f)(v \otimes w) &= g(f(v)w), \quad f,g \in V^*, \\ \theta_\ell(g \otimes f)(v \otimes w) &= f(vg(w)), \quad f,g \in {}^*V. \end{aligned} \tag{3.1}$$

Here $v, w \in V$. Both isomorphisms reverse the order of the factors.

Convention. *We identify $V^* \otimes_S V^*$ with $(V \otimes_S V)^*$ via θ_r, and ${}^*V \otimes_S {}^*V$ with ${}^*(V \otimes_S V)$ via θ_ℓ. Thus $R^\perp$ and ${}^\perp R$ in tensor algebras mean the inverse images of the orthogonal complements in Definition 2.39 under these evaluations.*

Definition 3.1. *The right quadratic dual and left quadratic dual of A are*

$$\begin{aligned} A^! &:= T_S(V^*)/(R^\perp), \quad R^\perp = \{F \in V^* \otimes_S V^* \mid \theta_r(F)(R) = 0\}, \\ {}^!A &:= T_S({}^*V)/({}^\perp R), \quad {}^\perp R = \{F \in {}^*V \otimes_S {}^*V \mid \theta_\ell(F)(R) = 0\}. \end{aligned} \tag{3.2}$$

We use $A^!$ unless a side is specified. The ordinary S-duals V^* and ${}^*V$ are placed in degree 1 in (3.2). By contrast, when V has degree 1, its graded S-dual $V^{(*)}$ has degree -1 by Definition 1.45.

By Proposition 2.20, a quadratic presentation identifies S with A_0, V with A_1, and R with $\operatorname{Ker}\mu_2$, where $\mu_2 : A_1 \otimes_{A_0} A_1 \to A_2$ is (2.6). The natural tensor-dual isomorphisms (2.3) therefore make the dual algebras independent of the chosen presentation up to isomorphism. An isomorphism *over S* means one restricting to id_S on degree 0.

> The constructions are those of [BGS96, §§2.7–2.8], with different star conventions: their V^* uses left S-linear maps and their ${}^*V$ uses right S-linear maps. Thus our $A^!$ is their ${}^!A$, and our ${}^!A$ is their $A^!$.

Proposition 3.2. *The algebra $A^!$ is quadratic and*

$$A^!_0 = S, \qquad A^!_1 = V^*, \qquad A^!_2 \cong R^*.$$

In particular, $\dim_{\Bbbk} A^!_2 = \dim_{\Bbbk} R$.

Proof Restriction to R gives an exact sequence of S-bimodules

$$0 \longrightarrow R^\perp \longrightarrow V^* \otimes_S V^* \xrightarrow{F \mapsto \theta_r(F)|_R} R^* \longrightarrow 0. \tag{3.3}$$

Surjectivity follows from Theorem 1.11: R has a complement as a right S-module, so a right S-linear map $R \to S$ extends by zero on that complement. Taking the quotient in (3.3) gives $A^!_2 \cong R^*$; Proposition 2.20 gives the remaining assertions about $A^!$.

> For the dimension assertion, Theorem 1.11 reduces to simple right modules over matrix rings. For a finite-dimensional division algebra Δ, the right $\mathrm{Mat}_n(\Delta)$-module $\Delta^{1\times n}$ has dual $\Delta^{n\times 1}$: a column c gives the homomorphism $x \mapsto cx$. The row and column spaces have the same $\Bbbk$-dimension, and dimensions add over simple summands. Hence $\dim_{\Bbbk} R^* = \dim_{\Bbbk} R$ without choosing a symmetrising form. $\square$

Adding relations has the opposite effect after dualising. For S-sub-bimodules $R \subset R' \subset V \otimes_S V$,

$$R \subset R' \quad \Longrightarrow \quad (R')^{\perp} \subset R^{\perp} \quad \Longrightarrow \quad \big(T_S(V)/(R')\big)^! \twoheadrightarrow \big(T_S(V)/(R)\big)^!.$$

The displayed surjection fixes S and is induced by id_{V^*} on the generators.

3.2 Recovering the original algebra

Lemma 3.3. *Let $U \subset M$ be an S-sub-bimodule of a finite-dimensional S-bimodule. Under the coevaluation isomorphisms of Lemma 1.46,*

$$^{\perp}(U^{\perp}) = U, \qquad (^{\perp}U)^{\perp} = U.$$

Proof For $u \in U$ and $f \in U^{\perp}$, coevaluation gives $\mathrm{coev}_M(u)(f) = f(u) = 0$, so $U \subset {}^{\perp}(U^{\perp})$ under Lemma 1.46. If $m \notin U$, Corollary 1.12 embeds M/U in $S^{\oplus n}$ as a right module for some n. A coordinate of the image of $m + U$ is non-zero. Composing $M \twoheadrightarrow M/U \hookrightarrow S^{\oplus n}$ with that coordinate projection gives a right S-linear map $f : M \to S$ satisfying $f(U) = 0$ and $f(m) \neq 0$. Thus $f \in U^{\perp}$ and $m \notin {}^{\perp}(U^{\perp})$. The proof of $(^{\perp}U)^{\perp} = U$ uses left modules instead. □

Theorem 3.4. *Left and right quadratic duality are canonically inverse:*

$$^!(A^!) \cong A \cong (^!A)^! \quad \textit{as graded algebras over } S.$$

Proof *Generators.* The coevaluation $\mathrm{coev}_V : V \xrightarrow{\sim} {}^*(V^*)$, $v \mapsto (f \mapsto f(v))$, of Lemma 1.46 identifies the generators of A and $^!(A^!)$. It preserves both S-actions by (2.2).

Relations. For $v, w \in V$ and $f, g \in V^*$, the same actions and (3.1) give

$$\begin{aligned} \theta_\ell(\mathrm{coev}_V(v) \otimes \mathrm{coev}_V(w))(g \otimes f) &= \mathrm{coev}_V(w)(gf(v)) = g(f(v)w) \\ &= \mathrm{coev}_{V \otimes_S V}(v \otimes w)(\theta_r(g \otimes f)). \end{aligned} \tag{3.4}$$

Thus $\mathrm{coev}_V \otimes \mathrm{coev}_V$ agrees with coevaluation on $V \otimes_S V$ under the tensor-dual identifications (2.3). Lemma 3.3, applied to $R \subset V \otimes_S V$, identifies the relations ${}^{\perp}(R^{\perp})$ of $^!(A^!)$ with R. The presentations (3.2) therefore give $^!(A^!) \cong A$ over S. Interchanging left and right gives $(^!A)^! \cong A$. □

Proposition 3.5. *A choice of symmetrising form $\tau : S \to \Bbbk$ gives isomorphisms $A^! \cong {}^!A$ and $(A^!)^! \cong A$ of graded algebras over S.*

Proof The maps $\alpha_V(f) = \tau \circ f$ and $\beta_V(g) = \tau \circ g$ in (1.13) give a bimodule isomorphism $\beta_V^{-1}\alpha_V : V^* \to {}^*V$. For $f, g \in V^*$, write $f' = \beta_V^{-1}\alpha_V(f)$ and $g' = \beta_V^{-1}\alpha_V(g)$. Then

$$\begin{aligned} \tau\big(f'(vg'(w))\big) = \tau\big(f(v)g'(w)\big) &= \tau\big(g'(f(v)w)\big) \\ = \tau\big(g(f(v)w)\big) \qquad (v, w \in V). & \end{aligned}$$

By (2.13), $(\beta_V^{-1}\alpha_V)^{\otimes_S 2}$ therefore identifies $R^{\perp}$ with ${}^{\perp}R$. The bimodule map $\beta_V^{-1} \circ \alpha_V$ extends by Proposition 2.16 to an isomorphism $T_S(V^*) \to T_S({}^*V)$ and hence descends to $A^! \cong {}^!A$. Apply the same comparison to $A^!$ and then Theorem 3.4 to obtain $(A^!)^! \cong {}^!(A^!) \cong A$. □

Theorem 3.4 needs no choice of form. Proposition 3.5 permits the usual statement that quadratic duality is an involution, but the identification of left and right duals uses τ.

3.3 Tensor algebras and monomial relations

Example 3.6. *Quadratic duality exchanges having no relations with setting every product of two generators equal to zero. Taking $R = 0$ or $R = V \otimes_S V$ in (3.2) gives*

$$\big(T_S(V)\big)^! = T_S(V^*)/(V^* \otimes_S V^*) \cong S \oplus V^*,$$
$$\big(T_S(V)/(V \otimes_S V)\big)^! = T_S(V^*).$$

In $S \oplus V^*$, $(V^*)^2 = 0$ and $(s, f)(t, g) = (st, sg + ft)$.

For the rest of this subsection, take $S = \bigoplus_{i=1}^r \Bbbk e_i$, with $e_i e_j = \delta_{ij} e_i$ and $\sum_i e_i = 1$. Choose a basis $\mathcal{B}_{ij}$ of each $e_i V e_j$ and let $\mathcal{B} = \bigcup_{i,j} \mathcal{B}_{ij}$. A non-empty word $b_1 \cdots b_d$ is composable if $b_t \in \mathcal{B}_{i_{t-1}, i_t}$ for $1 \le t \le d$ and some $i_0, \ldots, i_d \in \{1, \ldots, r\}$. Thus adjacent factors have matching local units. By Lemma 2.14, the composable words form a basis of the positive degree part of $T_S(V)$.

Definition 3.7. *The presentation $A = T_S(V)/(R)$ is monomial quadratic with respect to $\mathcal{B}$ if R is spanned by a subset $\mathcal{R}$ of the composable words of length 2 in $\mathcal{B}$.*

For $b \in \mathcal{B}_{ij}$, define the right S-linear map $b^* : V \to S$ by its values on $\mathcal{B}$:

$$b^*(c) = \delta_{bc} e_j \quad (c \in \mathcal{B}), \qquad e_j b^* e_i = b^*. \tag{3.5}$$

The elements b^* form a $\Bbbk$-basis of V^*; the left and right local units of each basis element are reversed.

Proposition 3.8. *For a monomial quadratic presentation:*

(1) *A basis of A consists of the e_i and the non-empty composable words containing no adjacent pair from $\mathcal{R}$.*

(2) *The quadratic dual is monomial, with*

$$R^\perp = \operatorname{span}_\Bbbk\{c^* \otimes_S b^* \mid bc \text{ is composable and } bc \notin \mathcal{R}\}. \tag{3.6}$$

Proof The ideal generated by $\mathcal{R}$ is spanned by the composable words containing an element of $\mathcal{R}$ as an adjacent pair. Removing those basis words proves (1). For $b \in \mathcal{B}_{ij}$ and $c \in \mathcal{B}_{j\ell}$, (3.1) and (3.5) give

$$\theta_r(c^* \otimes b^*)(b \otimes c) = c^*(e_j c) = e_\ell.$$

The functional vanishes on every other basis tensor. By Lemma 2.14, the composable $c^* \otimes b^*$ form a basis of $V^* \otimes_S V^*$. Evaluating a linear combination on each $b \otimes c \in \mathcal{R}$ forces precisely the coefficient of $c^* \otimes b^*$ to be zero, since $e_\ell \neq 0$. Thus the annihilator is the span of the complementary basis tensors, proving (2). □

Example 3.9. *We apply Example 3.6 to the tensor algebra description of $\Bbbk A_n$ in Example 2.21. Take $A = \Bbbk A_n$, $S = \bigoplus_{i=1}^n \Bbbk e_{ii}$, and $V = \bigoplus_{i=1}^{n-1} \Bbbk e_{i,i+1}$. There are no quadratic relations, so $A^!$ has the reversed generators $e^*_{i,i+1} \in e_{i+1,i+1} V^* e_{ii}$ and every composable product of two generators is zero. With C as in Example 2.33, this gives*

$$H_{A^!}(t) = I_n + tC^{\mathsf{T}}.$$

Here ^{T} denotes transpose: $(C^{\mathsf{T}})_{ij} = C_{ji}$. For $n = 3$,

$$(\Bbbk A_3)^! \cong \big(\Bbbk A_3/(e_{13})\big)^{\mathrm{op}}, \qquad \big(\Bbbk A_3/(e_{13})\big)^! \cong (\Bbbk A_3)^{\mathrm{op}}.$$

These are graded isomorphisms fixing the diagonal idempotents.

Example 3.10. *For $A = \Bbbk A_2$, Examples 2.33 and 3.9 give*

$$H_A(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}, \qquad H_{A^!}(-t)^{\mathsf{T}} = \begin{pmatrix} 1 & -t \\ 0 & 1 \end{pmatrix}, \qquad H_A(t) H_{A^!}(-t)^{\mathsf{T}} = I_2.$$

The transpose aligns the indices because an S-dual reverses the two idempotents in (3.5).

Example 3.11. *Proposition 3.8 exchanges the forbidden and surviving composable pairs. Take* $A = T_S(V)/(ab)$ *from Example 2.23, where* $S = \Bbbk e_1 \oplus \Bbbk e_2$, $e_1 a e_2 = a$, *and* $e_2 b e_1 = b$. *The only composable words of length* 2 *are* ab *and* ba. *Thus*

$$A^! = T_S(V^*)/(a^*b^*), \qquad A^! = S \oplus \Bbbk a^* \oplus \Bbbk b^* \oplus \Bbbk b^* a^*.$$

The original relation $ab = 0$ *corresponds to the surviving word* b^*a^*; *the surviving word* ba *corresponds to the dual relation* $a^*b^* = 0$. *In the order* e_1, e_2, *their Hilbert series are*

$$H_A(t) = \begin{pmatrix} 1 & t \\ t & 1+t^2 \end{pmatrix}, \qquad H_{A^!}(t) = \begin{pmatrix} 1+t^2 & t \\ t & 1 \end{pmatrix}.$$

The degree 2 *terms occur in different entries:* $ba \in e_2 A e_2$, *whereas* $b^*a^* \in e_1 A^! e_1$.

3.4 Symmetric and exterior algebras

Proposition 3.12. *For finite-dimensional* $\Bbbk$*-vector spaces* V, W,

$$\mathrm{Sym}(V)^! \cong \bigwedge D(V), \qquad \Big(\bigwedge W\Big)^! \cong \mathrm{Sym}(D(W)).$$

Proof For $\mathrm{Sym}(V)$, choose a basis $x_1, \dots, x_n$ of V, with dual basis $x_1^*, \dots, x_n^*$ of $D(V)$. Here $S = \Bbbk$, and the evaluation (3.1) reads

$$\theta_r(x_j^* \otimes x_i^*)(x_p \otimes x_q) = \delta_{ip}\delta_{jq}.$$

A tensor $\sum_{i,j=1}^n c_{ij} x_i^* \otimes x_j^*$ vanishes on each $x_p \otimes x_q - x_q \otimes x_p$ precisely when $c_{pq} = c_{qp}$ for $1 \le p < q \le n$. Thus the orthogonal complement of the symmetric relations is spanned by $(x_i^*)^2$ for $1 \le i \le n$ and $x_i^* x_j^* + x_j^* x_i^*$ for $1 \le i < j \le n$, which are the exterior relations on $D(V)$.

For $\bigwedge W$, choose a basis $y_1, \dots, y_m$ of W, with dual basis $y_1^*, \dots, y_m^*$ of $D(W)$. A tensor $\sum_{i,j=1}^m c_{ij} y_i^* \otimes y_j^*$ annihilates the exterior relations precisely when $c_{ii} = 0$ for $1 \le i \le m$ and $c_{pq} + c_{qp} = 0$ for $1 \le p < q \le m$. The orthogonal complement is therefore spanned by $y_i^* y_j^* - y_j^* y_i^*$ for $1 \le i < j \le m$, which are the symmetric relations on $D(W)$. The presentations in (2.7) give both isomorphisms. □

Consequently, to pair $\mathrm{Sym}(V)$ with $\bigwedge W$ by quadratic duality, take $W = D(V)$ and $y_i = x_i^*$, so $m = n$. Evaluation $D(D(V)) \cong V$ from Example 1.9 then identifies $(\bigwedge W)^!$ with $\mathrm{Sym}(V)$.

The proof works in characteristic 2: the relations $(x_i^*)^2 = 0$ remain part of the exterior presentation even when the signs $+$ and $-$ coincide. Example 2.40 is the calculation for two generators. Quadratic duality need not preserve Hilbert series: if $\dim_\Bbbk V > 0$, Example 2.35 gives different series for $\mathrm{Sym}(V)$ and $\mathrm{Sym}(V)^! \cong \bigwedge D(V)$.

The *Koszul condition*, to be defined in Lecture 5, implies the identity

$$H_A(t) H_{A^!}(-t)^{\mathsf{T}} = I_r \qquad \text{when } A_0 = \bigoplus_{i=1}^r \Bbbk e_i. \tag{3.7}$$

This is [BGS96, Lemma 2.11.1]; the proof will be given in Lecture 8. Equation (3.7) is not asserted for arbitrary quadratic algebras. For $\mathrm{Sym}(V)$ and its quadratic dual $\bigwedge D(V)$, with $n = \dim_\Bbbk V$, the case $S = \Bbbk$ gives $(1-t)^{-n}(1-t)^n = 1$.

3.5 Direct sums

For graded algebras A, B, the finite direct sum $A \oplus B$ has componentwise multiplication, unit $(1_A, 1_B)$, and $(A \oplus B)_i = A_i \oplus B_i$. In particular, its degree 0 algebra is $A_0 \oplus B_0$.

Proposition 3.13. *Let $A = T_S(V)/(R)$ and $B = T_T(W)/(Q)$ be quadratic algebras. Then $A \oplus B$ is quadratic over $S \oplus T$, and*

$$(A \oplus B)^! \cong A^! \oplus B^!.$$

Proof Let $(s,t) \in S \oplus T$ act on V through s and on W through t, on both sides. The local unit $(1_S, 0)$ is the identity on V and acts as zero on W, so

$$v \otimes_{S\oplus T} w = v(1_S,0) \otimes_{S\oplus T} w = v \otimes_{S\oplus T} (1_S,0)w = 0 \quad (v \in V,\ w \in W).$$

Likewise $w \otimes_{S\oplus T} v = 0$. Distributing tensor products over direct sums by Lemma 2.2(4) gives

$$T_{S\oplus T}(V \oplus W) \cong T_S(V) \oplus T_T(W),$$
$$(V \oplus W) \otimes_{S\oplus T} (V \oplus W) \cong (V \otimes_S V) \oplus (W \otimes_T W).$$

The relation space is $R \oplus Q$. A right $(S \oplus T)$-linear map $V \oplus W \to S \oplus T$ maps V into S and W into T, so the generator dual is $V^* \oplus W^*$. The annihilator of $R \oplus Q$ is therefore $R^\perp \oplus Q^\perp$, giving the assertion by Definition 3.1. □

For example, $\big((\Bbbk A_2)^{\oplus 2}\big)^! \cong \big((\Bbbk A_2)^{\mathrm{op}}\big)^{\oplus 2}$ by Example 3.9. Both degree 0 algebras are $\Bbbk^{\oplus 4}$; there are no products connecting the two summands.

3.6 Tensor products and signs

Definition 3.14. *For graded $\Bbbk$-algebras A, B, the ordinary tensor product $A \otimes_\Bbbk B$ and the graded tensor product $A \otimes_\Bbbk^{\mathrm{gr}} B$ have underlying vector space $A \otimes_\Bbbk B$ and the total grading of Definition 2.30, with products*

$$\begin{aligned} (a \otimes b)(a' \otimes b') &= aa' \otimes bb' && \text{in } A \otimes_\Bbbk B, \\ (a \otimes b)(a' \otimes b') &= (-1)^{\deg b \deg a'} aa' \otimes bb' && \text{in } A \otimes_\Bbbk^{\mathrm{gr}} B \end{aligned} \tag{3.8}$$

for homogeneous $a, a' \in A$ and $b, b' \in B$, extended bilinearly.

Both products have identity $1_A \otimes 1_B$. For the graded product, associativity follows from $\deg b \deg a' + (\deg b + \deg b') \deg a'' = \deg b' \deg a'' + \deg b(\deg a' + \deg a'')$ for homogeneous $a', a'' \in A$ and $b, b' \in B$. For homogeneous $a \in A$ and $b \in B$, the elements $a \otimes 1$ and $1 \otimes b$ commute in the ordinary tensor product; in the graded tensor product they satisfy

$$(1 \otimes b)(a \otimes 1) = (-1)^{\deg a \ \deg b}(a \otimes 1)(1 \otimes b).$$

Suppose $A_0 = S$ and $B_0 = T$ are finite-dimensional semisimple and $S \otimes_\Bbbk T$ is semisimple. The sign does not change the underlying graded $(S \otimes_\Bbbk T)$-bimodule, so the two tensor products have the same Hilbert series whenever A, B are locally finite. For $S = \Bbbk^{\oplus r}$ and $T = \Bbbk^{\oplus s}$, Lemma 2.36 gives its entries:

$$H_{A\otimes_\Bbbk B}(t)_{(i,p),(j,q)} = H_A(t)_{ij} H_B(t)_{pq}.$$

In particular, when $S = T = \Bbbk$, this is $H_A(t)H_B(t)$.

Proposition 3.15. *Let $A = T_S(V)/(R)$ and $B = T_T(W)/(Q)$ be quadratic algebras, and assume that $S \otimes_\Bbbk T$ is semisimple. Then both tensor products are quadratic over $S \otimes_\Bbbk T$, and*

$$(A \otimes_\Bbbk B)^! \cong A^! \otimes_\Bbbk^{\mathrm{gr}} B^!, \qquad (A \otimes_\Bbbk^{\mathrm{gr}} B)^! \cong A^! \otimes_\Bbbk B^!. \tag{3.9}$$

The mixed commuting relations become anticommuting relations under the reversed tensor pairing.

The assumption on $S \otimes_{\Bbbk} T$ holds when S, T are products of matrix algebras over $\Bbbk$, since $\mathrm{Mat}_a(\Bbbk) \otimes_{\Bbbk} \mathrm{Mat}_b(\Bbbk) \cong \mathrm{Mat}_{ab}(\Bbbk)$. It does not follow from semisimplicity alone: take $S = T$ to be the field in Example 1.40.

Proof The degree 1 part of $A \otimes_{\Bbbk} B$ is $(V \otimes_{\Bbbk} T) \oplus (S \otimes_{\Bbbk} W)$. By Lemma 2.2, its tensor square over $S \otimes_{\Bbbk} T$ decomposes into the four summands

$$(V \otimes_S V) \otimes_{\Bbbk} T \;\oplus\; \underbrace{V \otimes_{\Bbbk} W}_{\text{order } VW} \;\oplus\; \underbrace{V \otimes_{\Bbbk} W}_{\text{order } WV} \;\oplus\; S \otimes_{\Bbbk} (W \otimes_T W).$$

Its quadratic relations consist of $R \otimes_{\Bbbk} T$, $S \otimes_{\Bbbk} Q$, and the mixed relations

$$(v \otimes 1)(1 \otimes w) - (1 \otimes w)(v \otimes 1) \quad (v \in V,\ w \in W). \tag{3.10}$$

Here the products are taken in the tensor algebra on the degree 1 part, and their span is an $(S \otimes_{\Bbbk} T)$-sub-bimodule. Let C be the quotient of that tensor algebra by the three specified relation spaces. The generator maps induce algebra homomorphisms $A \to C$, $a \mapsto \overline{a}$, and $B \to C$, $b \mapsto \overline{b}$. Their images commute: the degree 0 actions already commute across the two factors, and (3.10) makes the degree 1 generators commute. Consequently $A \otimes_{\Bbbk} B \to C$, $a \otimes b \mapsto \overline{a}\,\overline{b}$, is an algebra homomorphism. It is inverse to the canonical homomorphism $C \to A \otimes_{\Bbbk} B$, since both composites fix the generating copies of S, T, V, W. Thus the stated relations give exactly $A \otimes_{\Bbbk} B$. For the graded tensor product, the same construction uses $\overline{b}\,\overline{a} = (-1)^{\deg a \deg b} \overline{a}\,\overline{b}$ for homogeneous a, b.

The right $(S \otimes_{\Bbbk} T)$-duals of the two generator summands are $V^* \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} W^*$, with evaluations

$$(f \otimes t)(v \otimes t') = f(v) \otimes tt', \qquad (s \otimes g)(s' \otimes w) = ss' \otimes g(w).$$

These identifications hold for finite free right modules and hence for their direct summands by Corollary 1.12. The reversed tensor evaluation pairs each pure summand with its corresponding dual summand and exchanges the two mixed summands; all other pairings are zero. The pure relation spaces have orthogonal complements $R^{\perp} \otimes_{\Bbbk} T$ and $S \otimes_{\Bbbk} Q^{\perp}$ in the corresponding pure summands. The mixed relation space is the set of pairs $(z, -z)$ in the two copies of $V \otimes_{\Bbbk} W$. For mixed tensors, the two reversed evaluations of $f \in V^*$ and $g \in W^*$ both give $f(v) \otimes g(w)$ on the corresponding copy. Thus an annihilating pair of functionals must have equal components. As these simple tensors span the full dual, the mixed annihilator is spanned by

$$(f \otimes 1)(1 \otimes g) + (1 \otimes g)(f \otimes 1) \quad (f \in V^*,\ g \in W^*).$$

These are the mixed relations of $A^! \otimes^{\mathrm{gr}}_{\Bbbk} B^!$, proving the first isomorphism in (3.9). For $A \otimes^{\mathrm{gr}}_{\Bbbk} B$, the minus sign in (3.10) is a plus sign. The mixed annihilator then has a minus sign, giving the ordinary tensor product on the dual side. □

Example 3.16. *We compute how Proposition 3.15 changes a commutation relation in a matrix example. Take $A = \Bbbk A_2 \otimes_{\Bbbk} \Bbbk A_2$. Its degree 0 algebra has basis $e_{ii} \otimes e_{jj}$ for $i, j \in \{1, 2\}$, and its degree 1 part has basis*

$$a = e_{12} \otimes e_{11}, \quad b = e_{22} \otimes e_{12}, \quad c = e_{11} \otimes e_{12}, \quad d = e_{12} \otimes e_{22}.$$

By Lemma 2.14, the tensor square of the degree 1 part has basis ab, cd, and both map to $e_{12} \otimes e_{12}$. Thus the relation space is $\Bbbk(ab - cd)$. Using the reversed generators from (3.5), its orthogonal complement is $\Bbbk(b^*a^* + d^*c^*)$. Hence the dual presentation has an anticommuting relation:

$$ab = cd \text{ in } A, \qquad b^*a^* = -d^*c^* \text{ in } A^!.$$

Order the degree 0 *idempotents as* $e_{11}\otimes e_{11}, e_{22}\otimes e_{11}, e_{11}\otimes e_{22}, e_{22}\otimes e_{22}$. *The homogeneous bases give*

$$H_A(t)=\begin{pmatrix}1 & t & t & t^2\\ 0 & 1 & 0 & t\\ 0 & 0 & 1 & t\\ 0 & 0 & 0 & 1\end{pmatrix}, \qquad H_{A^!}(t)=H_A(t)^{\mathsf{T}}.$$

The signs agree with Proposition 3.15; in characteristic 2, *the same presentations remain valid with* $-1=1$. *Here replacing* d^* *by* $-d^*$ *removes the minus sign. In this example the sign distinguishes the specified presentations, not the isomorphism classes of algebras.*

Example 3.17. *The symmetric/exterior pair shows that the sign in a graded tensor product cannot always be omitted. Let* $V=\Bbbk x_1\oplus\Bbbk x_2$ *and let* y_1, y_2 *be the dual basis of* $W=D(V)$. *Proposition 3.15 gives*

$$\mathrm{Sym}(V)^! \cong \big(\Bbbk[x_1]\otimes_{\Bbbk}\Bbbk[x_2]\big)^! \cong \frac{\Bbbk[y_1]}{(y_1^2)}\otimes_{\Bbbk}^{\mathrm{gr}}\frac{\Bbbk[y_2]}{(y_2^2)}\cong\bigwedge W.$$

If $\operatorname{char}\Bbbk\neq 2$, *the ordinary tensor product of the two square-zero algebras is commutative, whereas the graded tensor product is not:* $y_1y_2=-y_2y_1\neq 0$. *They are therefore not isomorphic. This sign cannot be removed by a change of generators.*

3.7 Exercises

Exercise 3.18. *For* $A=\Bbbk A_4/(e_{13})$, *write a presentation of* $A^!$ *using reversed generators and find homogeneous bases of* A *and* $A^!$. *Check the double dual using Proposition 3.8.* Hint. *Among* $e_{12}e_{23}$ *and* $e_{23}e_{34}$, *only the first is a relation. Complement the forbidden pairs and reverse their order.*

Exercise 3.19. *Let* $S=\Bbbk e_1\oplus\Bbbk e_2$ *and let* V *have basis* a, b *with* $e_1ae_2=a$ *and* $e_2be_1=b$. *Compute the quadratic dual of* $T_S(V)/(ab, ba)$ *and the Hilbert series of both algebras.* Check. *In the order* e_1, e_2, *the two series are*

$$\begin{pmatrix}1 & t\\ t & 1\end{pmatrix} \qquad \textit{and} \qquad \frac{1}{1-t^2}\begin{pmatrix}1 & t\\ t & 1\end{pmatrix}.$$

Exercise 3.20. *For the four generators in Example 3.16, replace the relation* $ab-cd$ *by* $ab-\lambda cd$, *where* $\lambda\in\Bbbk$. *Compute the quadratic dual relation for every* λ, *including* $\lambda=0$, *and recover the original relation by dualising again.* Hint. *Evaluate* $\mu b^*a^*+\nu d^*c^*$ *on* $ab-\lambda cd$. *For* $\lambda\neq 0$, *rescale* d *to compare with Example 3.16.*

4 Minimal graded resolutions

4.1 Projective covers and the graded Nakayama lemma

Let $A = \bigoplus_{d \geq 0} A_d$ be a locally finite graded algebra with $S = A_0$ finite-dimensional semisimple. Write

$$A_+ := \bigoplus_{d>0} A_d.$$

The grading makes A_+ a two-sided ideal, and projection onto degree 0 identifies A/A_+ with S. We regard S as a graded right A-module concentrated in degree 0, with action $s \cdot a = sa_0$ for $s \in S$ and $a = \sum_{d \geq 0} a_d \in A$.

All complexes below are chain complexes of graded right A-modules: their terms P_i are indexed by homological degrees $i \geq 0$, and their differentials are degree preserving maps $d_i : P_i \to P_{i-1}$ satisfying $d_{i-1}d_i = 0$ for $i \geq 2$. An augmented complex to a module M also has a degree preserving map $p_0 : P_0 \to M$ with $p_0 d_1 = 0$. The augmented complex is exact when $\operatorname{Im} d_{i+1} = \operatorname{Ker} d_i$ for $i \geq 1$, $\operatorname{Im} d_1 = \operatorname{Ker} p_0$, and p_0 is surjective.

For a graded right A-module M, define its *graded top* by

$$\overline{M} := M/MA_+.$$

The quotient $\overline{M}$ is a graded right S-module. We call M *bounded below* if $M_d = 0$ for all sufficiently small d. Every finitely generated graded module is bounded below: a finite generating set can be replaced by its finitely many homogeneous components, and A has no negative degrees.

Definition 4.1. *A graded right A-module P is projective if, for every surjective degree preserving map $q : X \to Y$ and every degree preserving map $f : P \to Y$, there is a degree preserving map $\widetilde{f} : P \to X$ such that $q\widetilde{f} = f$.*

The shifted free module $A(-d)$ has its generator 1 in degree d. If Q is a finite-dimensional projective right S-module, then $Q \otimes_S A(-d)$ is a projective graded right A-module: Q is a direct summand of a finite free right S-module, so $Q \otimes_S A(-d)$ is a direct summand of a finite direct sum of copies of $A(-d)$.

Lemma 4.2 (Graded Nakayama lemma). *If a bounded below graded right A-module M satisfies $M = MA_+$, then $M = 0$.*

Proof If $M \neq 0$, let d be the least degree with $M_d \neq 0$. Every degree d element of MA_+ is a sum of products $m_j a_{d-j}$ with $j < d$ and $d - j > 0$. All such m_j vanish by the choice of d, so $(MA_+)_d = 0$, contradicting $M_d = (MA_+)_d$. □

Definition 4.3. *A graded projective cover of a bounded below locally finite graded right A-module M is a surjective degree preserving map $p : P \to M$ from a bounded below locally finite graded projective module such that*

$$\overline{p} : P/PA_+ \xrightarrow{\sim} M/MA_+.$$

For a surjection $p : P \to M$, the condition in Definition 4.3 is equivalent to $\operatorname{Ker} p \subseteq PA_+$, since $p(PA_+) = MA_+$.

Proposition 4.4. *Every bounded below locally finite graded right A-module has a graded projective cover, unique up to isomorphism over M.*

Proof Put $\overline{M}_d = (M/MA_+)_d$. Since S is semisimple, the quotient map $M_d \to \overline{M}_d$ has a right S-linear section $s_d : \overline{M}_d \to M_d$. Set

$$P := \bigoplus_{d\in\mathbb{Z}} (\overline{M}_d \otimes_S A)(-d).$$

Each $\overline{M}_d$ is projective over S, so P is a graded projective right A-module. Since M is bounded below and locally finite, each degree of P is a finite sum of finite-dimensional spaces; hence P is locally finite. The maps

$$\overline{M}_d \otimes_S A \longrightarrow M, \qquad \overline{m} \otimes a \longmapsto s_d(\overline{m})a$$

define a degree preserving map $p : P \to M$. On graded tops, $\overline{p}$ identifies each copy of $\overline{M}_d$ with $(M/MA_+)_d$. Thus $\overline{p}$ is an isomorphism. The quotient $C := M/p(P)$ satisfies $C/CA_+ = 0$, hence $C = CA_+$; Lemma 4.2 gives $C = 0$. Therefore p is a projective cover.

For uniqueness, let $p : P \to M$ and $q : Q \to M$ be projective covers. Projectivity gives a map $f : P \to Q$ with $qf = p$. The maps $\overline{p}$ and $\overline{q}$ are isomorphisms, so $\overline{f} = \overline{q}^{-1}\overline{p}$ is an isomorphism. The quotient $Q/f(P)$ has zero graded top and is bounded below, hence is zero by Lemma 4.2. Thus f is surjective; since Q is projective, f splits. Its kernel is a direct summand of P, and its graded top is zero because $\overline{f}$ is injective. Lemma 4.2 gives $\operatorname{Ker} f = 0$, so f is an isomorphism over M. □

4.2 Minimal resolutions and graded Betti numbers

Definition 4.5. *A graded projective resolution of M is an exact sequence of degree preserving maps*

$$\cdots \longrightarrow P_2 \xrightarrow{d_2} P_1 \xrightarrow{d_1} P_0 \xrightarrow{p_0} M \longrightarrow 0$$

with every P_i graded projective. It is minimal if $p_0 : P_0 \to M$ is a graded projective cover and, for each $i \geq 1$, the map d_i with codomain restricted to $\operatorname{Ker} d_{i-1}$ is a graded projective cover, where $d_0 := p_0$.

Theorem 4.6. *Every bounded below locally finite graded right A-module has a minimal graded projective resolution. For any two such resolutions $P_\bullet \to M$ and $P'_\bullet \to M$, there are degree preserving isomorphisms $P_i \to P'_i$ commuting with the differentials and the maps to M.*

Proof Take a projective cover $p_0 : P_0 \to M$ using Proposition 4.4, and set $K_1 = \operatorname{Ker} p_0$. The module K_1 is bounded below and locally finite, so it has a projective cover $q_1 : P_1 \to K_1$. Repeat: after defining $q_i : P_i \to K_i$, set $K_{i+1} = \operatorname{Ker} q_i$ and cover K_{i+1} by $q_{i+1} : P_{i+1} \to K_{i+1}$. Composing each q_i with the inclusion $K_i \hookrightarrow P_{i-1}$ gives d_i and an exact minimal resolution.

For uniqueness, Proposition 4.4 identifies the covers of M in two minimal resolutions. The resulting isomorphism identifies their kernels; applying the same proposition to those kernels identifies the next projective covers. Induction gives an isomorphism in every homological degree, compatible with the differentials and the maps to M. □

Choose representatives $L_1, \ldots, L_r$ of the simple right S-modules, as in Definition 2.26. For a minimal resolution $P_\bullet \to M$, define the graded Betti numbers $\beta_{i,d,j}(M)$ by

$$(P_i/P_iA_+)_d \cong \bigoplus_{j=1}^{r} L_j^{\oplus \beta_{i,d,j}(M)}, \qquad \beta_{i,d}(M) := (\beta_{i,d,1}(M), \ldots, \beta_{i,d,r}(M)).$$

Here i is the homological degree, d is the grading degree, and j records the simple right S-module. Theorem 4.6 shows these multiplicities do not depend on the chosen minimal resolution. The projective cover construction also gives

$$P_i \cong \bigoplus_{d\in\mathbb{Z}} \bigoplus_{j=1}^{r} (L_j \otimes_S A)(-d)^{\oplus \beta_{i,d,j}(M)}.$$

For $S = \bigoplus_{j=1}^{r} \Bbbk e_j$, the simple modules are $L_j = e_j S$ and $L_j \otimes_S A \cong e_j A$.

Example 4.7. *This example shows how a non-field degree zero algebra gives distinct projective covers for its different simple modules. Let $A = \Bbbk A_2$, with $S = \Bbbk e_{11} \oplus \Bbbk e_{22}$ and $A_1 = \Bbbk e_{12}$. The simple right S-modules are $L_1 = e_{11}S$ and $L_2 = e_{22}S$, with projective lifts $Q_1 = e_{11}A$ and $Q_2 = e_{22}A$. The quotient map $Q_1 \to L_1$ has kernel $\Bbbk e_{12}$, while $Q_2 = L_2$. Sending the degree 1 generator e_{22} to e_{12} gives the minimal resolution*

$$0 \longrightarrow Q_2(-1) \xrightarrow{d_1} Q_1 \longrightarrow L_1 \longrightarrow 0, \qquad d_1(e_{22}) = e_{12}.$$

Example 4.8. *For $A = \Bbbk[x]$, with $\deg x = 1$, the degree zero quotient module is $\Bbbk = A/(x)$. Multiplication by x gives the minimal resolution*

$$0 \longrightarrow A(-1) \xrightarrow{d_1} A \xrightarrow{\pi} \Bbbk \longrightarrow 0, \qquad d_1(a) = xa, \quad \pi(f) = f(0).$$

The map d_1 has degree 0, is injective, and has image $xA = \operatorname{Ker} \pi \subset AA_+$. Thus $\beta_{0,0} = 1$ and $\beta_{1,1} = 1$.

Example 4.9. *The exterior algebra on one generator gives an infinite minimal resolution, although the algebra itself is finite-dimensional. Let $E = \Bbbk[\epsilon]/(\epsilon^2)$ with $\deg \epsilon = 1$ and $E_+ = \Bbbk\epsilon$. The degree zero quotient is $\Bbbk = E/E_+$, and*

$$\cdots \xrightarrow{d_3} E(-2) \xrightarrow{d_2} E(-1) \xrightarrow{d_1} E \xrightarrow{\pi} \Bbbk \longrightarrow 0, \qquad d_i(a) = \epsilon a, \quad \pi(a) = a_0.$$

Each d_i is degree preserving and right E-linear. Since $\operatorname{Ker}(\epsilon \cdot -) = \epsilon E = \operatorname{Im}(\epsilon \cdot -)$, the sequence is exact at every $E(-i)$ and at E. Every differential has image in the preceding term times E_+, so the resolution is minimal and $\beta_{i,i} = 1$ for all $i \geq 0$.

If M is finitely generated, its first projective cover is finitely generated. Without a noetherian hypothesis on A, later kernels need not be finitely generated, so later terms of the minimal resolution need not be finitely generated. The bounded below locally finite setting above still gives projective covers and a minimal resolution.

5 Koszul complex and Koszul algebras

5.1 The Koszul complex of a polynomial algebra

Let V be an n-dimensional $\Bbbk$-vector space, choose a basis $x_1, \dots, x_n$, and let $A = \mathrm{Sym}(V) = \Bbbk[x_1, \dots, x_n]$, with each x_j in degree 1. Let $\pi : A \to \Bbbk$ be projection onto degree 0, so $\Bbbk = A/A_+$. For $0 \le i \le n$, put

$$K_i := \bigwedge^i V \otimes_{\Bbbk} A(-i), \qquad K_0 = A.$$

The right A-action is on the second factor. The factor $\bigwedge^i V$ is placed in grading degree 0, so the generators of K_i have grading degree i.

Definition 5.1. *For $1 \le i \le n$, define the Koszul differential*

$$\begin{aligned} d_i : K_i &\longrightarrow K_{i-1}, \\ (v_1 \wedge \cdots \wedge v_i) \otimes a &\longmapsto \sum_{r=1}^{i} (-1)^{r-1} (v_1 \wedge \cdots \wedge \widehat{v_r} \wedge \cdots \wedge v_i) \otimes v_r a. \end{aligned} \tag{5.1}$$

The formula is multilinear and alternating in $v_1, \dots, v_i$, so it defines a map on $\bigwedge^i V$. The augmented chain complex $K_\bullet \to \Bbbk$ consists of these maps and $\pi : K_0 = A \to \Bbbk$.

Proposition 5.2. *The augmented complex*

$$0 \longrightarrow K_n \xrightarrow{d_n} \cdots \xrightarrow{d_2} K_1 \xrightarrow{d_1} K_0 \xrightarrow{\pi} \Bbbk \longrightarrow 0$$

is an exact minimal graded free resolution of $\Bbbk$ over A.

Proof Each d_i preserves degree and is right A-linear. For $i \ge 2$, compute $d_{i-1}d_i$ on a basis wedge. Every term is indexed by two distinct positions $r < s$ deleted from that wedge. Deleting position r and then s gives coefficient $(-1)^{r+s-3} x_{j_s} x_{j_r}$; deleting s and then r gives coefficient $(-1)^{r+s-2} x_{j_r} x_{j_s}$. Since $x_{j_s} x_{j_r} = x_{j_r} x_{j_s}$, these two terms sum to zero. Hence $d_{i-1}d_i = 0$; also $\pi d_1 = 0$ because every term of d_1 is divisible by a variable.

To prove exactness, give A and $\bigwedge V$ the multigrading with $\deg x_j$ equal to the jth standard basis vector. For $\alpha \in \mathbb{Z}^n_{\ge 0}$, let $\mathrm{supp}(\alpha) = \{j \mid \alpha_j > 0\}$. The degree α part of K_i has basis

$$b_J := (x_{j_1} \wedge \cdots \wedge x_{j_i}) \otimes x^{\alpha - \mathbf{1}_J}, \qquad J = \{j_1 < \cdots < j_i\} \subseteq \mathrm{supp}(\alpha),$$

where $\mathbf{1}_J$ has entry 1 at indices in J and 0 elsewhere. The differential is $d_i(b_J) = \sum_{r=1}^{i} (-1)^{r-1} b_{J \setminus \{j_r\}}$.

If $\alpha \ne 0$, let q be the least index in $\mathrm{supp}(\alpha)$. For each $i \ge 0$, define $h_{\alpha,i} : K_{i,\alpha} \to K_{i+1,\alpha}$ on basis vectors by

$$h_{\alpha,i}(b_J) = \begin{cases} b_{\{q\} \cup J}, & q \notin J, \\ 0, & q \in J, \end{cases}$$

inserting q first in the ordered wedge. For $q \notin J$ and $i \ge 1$, direct substitution gives

$$\begin{aligned} d_{i+1} h_{\alpha,i}(b_J) &= b_J - \sum_{r=1}^{i} (-1)^{r-1} b_{\{q\} \cup (J \setminus \{j_r\})}, \\ h_{\alpha,i-1} d_i(b_J) &= \sum_{r=1}^{i} (-1)^{r-1} b_{\{q\} \cup (J \setminus \{j_r\})}. \end{aligned}$$

For $i = 0$, $J = \varnothing$ and $d_1 h_{\alpha,0}(b_\varnothing) = b_\varnothing$. For $q \in J$, the term of $d_i(b_J)$ deleting q is the only term on which $h_{\alpha,i-1}$ is nonzero, and it gives $h_{\alpha,i-1} d_i(b_J) = b_J$ while $d_{i+1} h_{\alpha,i}(b_J) = 0$. Thus $d_{i+1} h_{\alpha,i} + h_{\alpha,i-1} d_i =$

$\mathrm{id}_{K_{i,\alpha}}$ for every $i \geq 1$ in positive multidegree; the case $i = 0$ was checked separately. All positive multidegrees are therefore exact. In multidegree 0, only $K_0 \cong \Bbbk$ is nonzero and π is the identity onto $\Bbbk$. This proves exactness of the augmented complex.

Finally, each K_i is a finite direct sum of copies of $A(-i)$, and (5.1) has image in $K_{i-1}A_+$. Exactness gives $\operatorname{Ker} d_i = \operatorname{Im} d_{i+1} \subseteq K_i A_+$ for $i < n$, and $\operatorname{Ker} d_n = 0$. Also, $\operatorname{Ker} \pi = AA_+$. Thus each map from K_i onto the preceding kernel has kernel contained in $K_i A_+$, so it is a projective cover by Definition 4.3. The resolution is therefore minimal. □

5.2 Koszul algebras

Definition 5.3. *A minimal graded projective resolution $P_\bullet \to M$ is linear if P_i/P_iA_+ is concentrated in degree i for every $i \geq 0$. This is equivalent to P_i being generated in degree i: generation in degree i implies the stated concentration because A_+ raises degree. Conversely, if the top is concentrated in degree i, boundedness below and Lemma 4.2 show that P_i has no components below degree i; the quotient $P_i/(P_i)_iA$ then has zero graded top, so Lemma 4.2 makes that quotient zero.*

Definition 5.4. *Let $A = T_S(V)/(R)$ be a quadratic algebra with $S = A_0$ finite-dimensional semisimple. The algebra A is Koszul if the degree zero quotient module $S = A/A_+$ has a linear minimal graded projective resolution as a right A-module.*

The generators in homological degree i must then all have grading degree i. Since S can have several simple modules, the definition requires a linear resolution of their direct sum, not just of one chosen simple module. For a minimal resolution, Definition 5.3 is equivalently the vanishing $\beta_{i,d,j}(S) = 0$ whenever $d \neq i$.

Proposition 5.5. *For every finite-dimensional vector space V, the symmetric algebra $\operatorname{Sym}(V)$ is Koszul.*

Proof Proposition 5.2 gives a minimal resolution with $K_i = \bigwedge^i V \otimes_\Bbbk A(-i)$. Thus K_i is generated in degree i for every i. □

Example 5.6. *The definition also applies when S has several simple modules. For $A = \Bbbk A_2$, the resolution in Example 4.7 is linear for L_1, and $L_2 = e_{22}A$ is projective in degree 0. Taking their direct sum gives a linear resolution of $S = L_1 \oplus L_2$; hence $\Bbbk A_2$ is Koszul.*

Example 5.7. *The exterior algebra $E = \Bbbk[\epsilon]/(\epsilon^2)$ is quadratic, and Example 4.9 gives a minimal resolution of its degree zero quotient with $P_i = E(-i)$. Thus E is Koszul, although its minimal resolution is infinite.*

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